RESULTS

4.0 Introduction

This chapter presents the analysis and findings of the study on the lived experiences of women living with obstetric fistula in Aisha Buhari Mother and Child Hospital, Ilorin, Kwara State. The purpose of this chapter is to systematically present the qualitative data generated from in-depth interviews and to organize the findings in a manner that addresses the research objectives and questions guiding the study. A qualitative phenomenological approach was employed to gain deep insight into the physical, emotional, social, and coping experiences of the participants. The data were analyzed thematically, allowing patterns, categories, and themes to emerge from the participants' accounts. This systematic approach ensured that the findings remained grounded in the participants' voices while maintaining alignment with the study objectives. The findings are presented according to major themes and sub-themes derived from the data, including demographic characteristics, physical experiences, emotional experiences, social experiences, and coping strategies. Participants' verbatim quotations are incorporated to enhance credibility and to preserve the authenticity of their narratives. Where necessary, tables are used to summarize thematic categories and provide clarity. This chapter therefore provides a comprehensive presentation of the findings, forming the basis for subsequent discussion, interpretation, and recommendations in the following chapter.

Table 4.1: Showing Socio-demographic Characteristics of the Participants (n = 22)

S/N	Age	Gender	Marital Status	Educational Qualification	Occupation	Number Of Children	Religion	Ethnic Group
P1	20–29	Female	Married	Tertiary	Civil Servant	1	Christianity	Yoruba
P2	30–39	Female	Married	Secondary	Self-employed	3	Christianity	Igbo
P3	30+	Female	Widowed	No formal education	Trader	>5	Islam	Yoruba
P4	20–29	Female	Married	Tertiary	Private sector	1	Christianity	Yoruba
P5	20–29	Female	Married	Tertiary	Civil Servant	2	Islam	Hausa
P6	<20	Female	Single	Secondary	Unemployed	1	Christianity	Yoruba
P7	20–29	Female	Married	Tertiary	Civil Servant	1	Christianity	Yoruba
P8	20–29	Female	Married	Tertiary	Civil Servant	1	Christianity	Yoruba
P9	30–39	Female	Married	Secondary	Self-employed	2–3	Islam	Hausa
P10	35+	Female	Married	Primary	Self-employed	5	Islam	Yoruba
P11	20–29	Female	Married	Tertiary	Civil Servant	2	Christianity	Yoruba
P12	26	Female	Married	Tertiary	Civil Servant	1	Christianity	Yoruba
P13	<20	Female	Single	Secondary	Unemployed	1	Christianity	Yoruba
P14	30–39	Female	Married	Secondary	Self-employed	3	Christianity	Igbo
P15	30+	Female	Widowed	No formal education	Trader	>5	Islam	Yoruba
P16	20–29	Female	Married	Tertiary	Private sector	1	Christianity	Yoruba
P17	20–29	Female	Married	Tertiary	Civil Servant	2	Islam	Hausa
P18	<20	Female	Single	Secondary	Unemployed	1	Christianity	Yoruba
P19	20–29	Female	Married	Tertiary	Civil Servant	1	Christianity	Yoruba
P20	30–39	Female	Married	Secondary	Self-employed	2–3	Islam	Hausa
P21	35+	Female	Married	Primary	Self-employed	5	Islam	Yoruba
P22	20–29	Female	Married	Tertiary	Civil Servant	2	Ch	

4.1 The availability of blood loss measurement tools in postpartum haemorrhage detection among health workers in Adeoyo Maternity Teaching Hospital Yemetu, Ibadan

Table 4.2: Themes and Sub-themes on Availability of Blood Loss Measurement Tools

Theme	Sub-theme
1. Non -availability of Blood drapes for accurate measurement	1.1 Lack of blood drapes for accurate measurement 1.2 Use of visual estimation when tools are unavailable

THEME 1: Non-availability of Blood Drapes for Accurate Measurement

Many participants expressed the challenge of non-availability of blood drapes, a crucial tool for accurately measuring blood loss in postpartum hemorrhage (PPH) cases. Due to the lack of this equipment, healthcare workers often had to resort to visual estimation or other methods, which were not as precise or reliable. The non-availability of blood drapes was a significant barrier in ensuring the accuracy of measurements during deliveries, particularly Cesarean sections (CS) or high-risk births. Below are the descriptions of the challenges faced by participants in the absence of blood drapes for blood loss measurement:

Subtheme 1.1: Lack of Blood Drapes for Accurate Measurement

The lack of blood drapes was consistently highlighted as a critical challenge. Respondents noted that blood drapes, which are essential for calibrated blood loss measurement, were

either unavailable or in short supply. Without blood drapes, healthcare workers were forced to rely on visual estimation, which lacks the precision needed for accurate clinical intervention. The absence of this tool contributed to the inaccuracy of blood loss measurement, which could potentially delay timely interventions for postpartum hemorrhage cases.

"We don't have blood drapes in this facility, and that affects our ability to measure blood loss accurately. Without them, we are left to estimate." (Respondent 3, Nurse, 13 years)

"The absence of blood drapes means we have to guess the amount of blood loss, and it's frustrating knowing that it's not precise." (Respondent 5, Nurse-Midwife, 16 years)

"In the absence of drapes, I rely on gauze or visual estimation to make a judgment. This is far from ideal." (Respondent 8, Midwife, 6 years)

"Without drapes, we can't get an accurate reading of blood loss. Sometimes we use gauze, but it's not as effective." (Respondent 2, Nurse, 7 years)

"Blood drapes are essential for accuracy, but sadly, they are not available in our facility." (Respondent 10, Nurse-Midwife, 5 years)

"We don't use drapes at all. If we had them, the measurement would be far more accurate." (Respondent 12, Midwife, 8 years)

"Having blood drapes would make a huge difference, but at the moment, they're just not available in the hospital." (Respondent 7, Nurse-Midwife, 6 years)

Subtheme 1.2: Use of Visual Estimation When Tools Are Unavailable

When blood drapes were not available, healthcare workers resorted to visual estimation as a substitute method for measuring blood loss. Although visual estimation can provide a rough idea of the blood loss, it lacks the accuracy required for timely clinical decisions. Many participants expressed frustration with having to rely on this method, as it could lead to missed diagnoses or delayed treatment. Visual estimation was especially used in emergency situations where resources were limited.

"When we don't have the drapes, we just estimate by looking at the blood on the gauze or pads. It's not accurate, but it's the only option." (Respondent 4, Nurse, 5 years)

"I use visual estimation mostly when tools are unavailable. Sometimes, I have to make quick decisions, and estimation is all I have." (Respondent 6, Nurse-Midwife, 12 years)

"With visual estimation, I have to guess the volume of blood lost. It's not a great feeling knowing that we're not measuring accurately." (Respondent 3, Midwife, 8 years)

"I was trained to use calibrated tools, but without them, visual estimation is the fallback. It's not as good as using the right equipment, but it works in some cases." (Respondent 9, Nurse, 6 years)

"Visual estimation is what we rely on when we don't have the tools, but it's far less reliable than the proper equipment." (Respondent 5, Midwife, 7 years)

"Without drapes, we end up just looking at the blood and guessing how much was lost. It's not perfect, and I always wonder if I missed something." (Respondent 8, Nurse, 10 years)

"There's a lot of guesswork when it comes to measuring blood loss. Visual estimation is all we have when we don't have blood drapes." (Respondent 10, Midwife, 4 years)

4.2 The use of blood loss measurement tools in postpartum haemorrhage detection among health workers in Adeoyo Maternity Teaching Hospital Yemetu, Ibadan

Table 4.3: Themes and Sub-themes on Use of Blood Loss Measurement Tools

Theme	Sub-theme
1. Regular Use of Available Tools	1.1 Use of gauze for blood loss measurement 1.2 Use of suction machines and calibrated jars for PPH detection
2. Ad-hoc Use and Reliance on Visual Estimation	2.1 Use of visual estimation when other tools are not in use 2.2 Estimation during high-patient volume or emergencies
3. Factors Influencing the Use of Tools	3.1 Influence of training and experience on tool usage 3.2 Availability of tools influencing frequency of use
4. Inconsistent Use Across Different Settings	4.1 Variations in tool use between different wards or units 4.2 Differences in use during Cesarean vs spontaneous vaginal deliveries

THEME 1: Regular Use of Available Tools

Many respondents highlighted that blood loss measurement tools such as gauze, suction machines, and calibrated jars were regularly used in their facilities to detect postpartum

hemorrhage (PPH). These tools were often employed in situations where postpartum hemorrhage was suspected, and the tools were available in most cases. The respondents' descriptions indicate that when tools were available, they were actively used to measure blood loss accurately, contributing to the timely management of PPH. Below are the descriptions of the regular use of tools:

Subtheme 1.1: Use of Gauze for Blood Loss Measurement

In many cases, respondents mentioned the use of gauze to measure blood loss. This was particularly useful in spontaneous vaginal deliveries (SVDs), where gauze would be weighed before and after use to estimate the volume of blood lost. The standardized weight of a soaked gauze (e.g., 20 mL per gauze) was often cited as a means to approximate blood loss. The routine use of gauze as a blood loss measurement tool was mentioned across multiple responses, particularly for deliveries where bleeding was less severe.

"We use gauze to estimate blood loss. A fully soaked gauze is about 20 mL of blood, so we just weigh the gauze before and after use." (Respondent 4, Nurse, 5 years)

"During vaginal deliveries, gauze is the most used tool. We weigh it before and after to know how much blood is lost." (Respondent 2, Midwife, 6 years)

"Gauze is the go-to tool for measuring blood loss in spontaneous vaginal deliveries. We can rely on it to get an approximate amount of blood lost." (Respondent 9, Nurse, 8 years)

"In cases where the delivery is not complicated, gauze is used to estimate blood loss. It's quick and easy, and we have it available all the time." (Respondent 7, Midwife, 4 years)

"For me, gauze is the most accessible tool for estimating blood loss during normal deliveries. I use it all the time, and it's quite reliable for most cases." (Respondent 3, Nurse, 13 years)

"When blood loss is minimal, gauze works well. A saturated gauze gives us a decent estimate of blood loss." (Respondent 5, Nurse, 6 years)

"The gauze method is simple, but effective for spontaneous vaginal deliveries. We rely on it for quick estimates." (Respondent 6, Midwife, 9 years)

"The gauze is a standard tool for us in estimating blood loss in non-cesarean deliveries. It's fast and helps us manage patients more efficiently." (Respondent 10, Nurse, 12 years)

Subtheme 1.2: Use of Suction Machines and Calibrated Jars for PPH Detection

For Cesarean sections (CS) and more severe cases of PPH, suction machines and calibrated jars were commonly used to quantify blood loss. These tools provided more precise measurements compared to gauze and visual estimation, which is why they were preferred when available. Respondents frequently mentioned using suction machines to remove blood during surgery and then measuring the blood in calibrated jars. The accuracy of these tools was highlighted as being particularly important in managing larger blood losses that required more precise monitoring.

"In Cesarean sections, we use suction machines to remove the blood, and then we measure the blood in calibrated jars. This is the most accurate method we have." (Respondent 7, Surgeon, 8 years)

"During surgery, we use suction to remove blood and then measure it in calibrated jars. This tool gives us a good idea of how much blood has been lost." (Respondent 9, Nurse-Midwife, 6 years)

"For higher volumes of blood loss, especially during Cesarean sections, we always use calibrated jars and suction machines. These tools are key to accurate measurement." (Respondent 5, Nurse, 7 years)

"Suction machines and calibrated jars are essential for managing blood loss in Cesarean deliveries. I use them regularly when performing these procedures." (Respondent 12, Midwife, 8 years)

"Calibrated jars are my go-to tool for measuring blood loss during surgical deliveries. They provide an accurate count, which is crucial for deciding on interventions." (Respondent 3, Nurse, 9 years)

"For Cesarean sections, suction machines give us an immediate measurement, and the calibrated jars allow us to precisely calculate the blood loss." (Respondent 8, Nurse, 6 years)

"We use calibrated jars in surgeries to measure blood loss directly. It's the most reliable method, especially when there's significant bleeding." (Respondent 4, Surgeon, 5 years)

"During Cesarean sections, the suction machine is indispensable. We immediately collect blood in calibrated jars and ensure that we accurately measure the blood loss." (Respondent 6, Midwife, 7 years)

THEME 2: Ad-hoc Use and Reliance on Visual Estimation

While healthcare facilities are equipped with tools such as gauze, suction machines, and calibrated jars for measuring blood loss in postpartum hemorrhage (PPH) detection, many healthcare workers reported that these tools were not always readily available. In such cases, visual estimation became a necessary and common practice. This reliance on visual estimation often occurred during emergencies or in high-patient volume situations, where time constraints or a lack of resources prevented the use of more accurate measurement tools. Though visual estimation is subjective and less accurate, it was widely used as a quick and immediate alternative in resource-limited settings.

Subtheme 2.1: Use of Visual Estimation When Other Tools Are Not in Use

Respondents frequently mentioned that when more accurate tools like calibrated jars or suction machines were unavailable, they had no choice but to estimate blood loss visually. This approach was often used when resources were scarce, and in situations requiring immediate action to manage PPH, where waiting for a tool to become available could be dangerous for the patient. While this method lacked the precision of other tools, it was considered better than nothing in such critical moments.

"When the tools are not available, we just estimate visually. I know it's not perfect, but it's the fastest way to gauge blood loss." (Respondent 8, Nurse, 7 years)

"If we don't have the calibrated jars or suction machines, we rely on visual estimation. We estimate based on the amount of blood on the pad or gauze." (Respondent 10, Midwife, 5 years)

"Sometimes, we don't have the suctioning machine, and that's when I just estimate the blood loss. It's not ideal, but it's all we can do." (Respondent 6, Nurse, 12 years)

"When there is no calibrated jar available, I just look at the amount of blood soaked in the gauze and estimate the volume. It's a quick method but not always accurate." (Respondent 7, Nurse, 8 years)

"We just use visual estimation during emergencies. Sometimes we can't wait for tools to be brought in, so we estimate how much blood is lost based on what we see." (Respondent 5, Nurse-Midwife, 6 years)

"When there's a shortage of suction machines, we visually estimate blood loss. We look at how much blood has been lost on the mat or gauze." (Respondent 2, Nurse, 13 years)

"It's always frustrating when we don't have the proper tools, but when that happens, we have to rely on our judgment and estimate the blood loss visually." (Respondent 3, Nurse, 10 years)

"Visual estimation is our fallback option when the necessary tools aren't available. It's a bit subjective, but it helps us make decisions quickly." (Respondent 9, Midwife, 6 years)

Subtheme 2.2: Estimation During High-Patient Volume or Emergencies

High-patient volume and emergencies were also significant factors that led to the increased reliance on visual estimation. Many respondents noted that during periods of overcrowding or high demand, such as in busy wards or theaters with numerous patients, it became difficult to accurately measure blood loss using calibrated tools. In such cases, quick decisions had to be made, and visual estimation was often the only option. Respondents indicated that in these settings, the subjectivity of visual estimation was sometimes accepted as a practical necessity due to the urgency of managing PPH.

"When the ward is really busy, and we have a lot of patients, we sometimes don't have time to properly measure blood loss, so we just estimate based on what we see." (Respondent 4, Nurse, 6 years)

"During emergencies, we have to act fast, and sometimes we just estimate the blood loss because we don't have time to measure it properly." (Respondent 7, Nurse, 8 years)

"When there are too many patients, it becomes impossible to use the tools. So, we rely on visual estimation to keep things moving." (Respondent 10, Nurse-Midwife, 6 years)

"When the hospital is understaffed and we have many patients in labor, it's hard to measure blood loss properly, so we just look at the pads or gauze and estimate." (Respondent 6, Midwife, 9 years)

"In the busy surgical ward, we don't always have the suction machine available, so we estimate the blood loss as best we can from what we see." (Respondent 5, Nurse, 6 years)

"With high-patient volume, the tools get used up quickly, and we can't always get the precise measurements, so we visually estimate." (Respondent 8, Nurse, 5 years)

"We don't always have calibrated jars or suction machines during emergencies, so we rely on estimating the blood loss from visual observation." (Respondent 3, Midwife, 10 years)

"Sometimes we just have to estimate the blood loss because we don't have the proper tools during high-demand situations." (Respondent 9, Nurse, 7 years)

THEME 3: Lack of Tools and Equipment for Accurate Blood Loss Measurement

Lack of adequate tools and equipment limitations emerged as significant barriers to accurate blood loss measurement in many healthcare facilities. Several respondents indicated that although certain measurement tools, like gauze and suction machines, were available, there was a consistent shortage or inadequate functioning of critical equipment. Specifically, the lack of blood drapes and calibrated jars made it difficult for healthcare workers to perform accurate blood loss measurements in postpartum hemorrhage (PPH) cases. These equipment deficiencies were especially pronounced in resource-limited settings, leading to reliance on visual estimation or improvised methods that compromised the quality of care provided.

Subtheme 3.1: Insufficient Quantity of Key Equipment

Many respondents highlighted that while some tools for blood loss measurement were available, there were often not enough of them to meet the demand. For instance, suction machines and calibrated jars were often mentioned as critical tools, but healthcare workers often reported having only a few functional units, which were inadequate for handling multiple patients in high-traffic wards or operating theaters. The insufficient quantity of these tools was identified as a significant constraint, especially in emergency situations where quick and accurate measurements are necessary.

"We have only one suction machine in the entire ward. When it's in use, we have to wait until it's free, and in that time, we rely on visual estimation." (Respondent 4, Nurse, 7 years)

"The calibrated jars we have are limited, so we use them for only the most critical cases. For other deliveries, we have to estimate the blood loss." (Respondent 6, Midwife, 10 years)

"Sometimes, we only have one suction machine for a whole department. It's not enough to handle the number of patients we have, so we have to make do with visual estimation." (Respondent 2, Nurse, 8 years)

"In the emergency room, we only have a few calibrated jars. It's always difficult to get accurate measurements when we don't have enough." (Respondent 5, Nurse, 12 years)

"One suction machine is not enough for a busy department, especially during surgeries. We need more, but it's not always available." (Respondent 9, Midwife, 6 years)

"There are not enough calibrated jars to go around. We have to improvise and use what we have, but it doesn't always give us an accurate measurement." (Respondent 7, Nurse, 10 years)

"Our department doesn't have enough calibrated jars or suction machines, so we often resort to visual estimation, which is less reliable." (Respondent 3, Nurse, 6 years)

"We always face a shortage of blood measurement tools. Sometimes, only one calibrated jar is available, and it's not enough for multiple patients." (Respondent 8, Nurse, 6 years)

Subtheme 3.2: Unavailability of Critical Tools Like Blood Drapes

A major limitation mentioned by several respondents was the unavailability of blood drapes, a tool that is critical for accurately measuring blood loss. While suction machines and calibrated jars are commonly used in many facilities, the absence of blood drapes made it harder to measure the blood loss in a controlled and precise manner. Respondents expressed that blood drapes were rarely available in their facilities, and the lack of this tool led to improvised methods, such as gauze weight estimation or visual inspection, to estimate the blood loss. This limitation was particularly evident in high-risk cases where accurate measurement was crucial.

> *"We don't have blood drapes here. If we had them, it would be easier to measure blood loss during surgery. Instead, we use gauze and try to estimate the loss."* (Respondent 6, Nurse-Midwife, 5 years)
>
> *"Blood drapes are essential for accurate measurement, but we don't have them in our facility. Without them, we have to guess the amount of blood lost."* (Respondent 7, Midwife, 8 years)
>
> *"I've heard of blood drapes being used in other facilities, but here, we don't have them. We just use gauze or visual estimation."* (Respondent 3, Nurse, 10 years)

"We lack blood drapes, which makes it harder to get an accurate measurement. We try to make do with gauze and suction machines, but it's not the same." (Respondent 4, Nurse, 7 years)

"The absence of blood drapes is a significant issue in our facility. We don't have them, so we use other methods, but they aren't as reliable." (Respondent 8, Nurse, 5 years)

"Blood drapes are not available here. I've only seen them in a few other hospitals. We manage without them, but it means we don't get precise measurements." (Respondent 2, Midwife, 6 years)

"In our facility, blood drapes are not something we have. If they were available, it would improve how we measure blood loss in postpartum hemorrhage." (Respondent 6, Nurse, 5 years)

"We don't have any blood drapes here, which limits our ability to measure blood loss accurately. We're forced to use whatever we have, but it's not ideal." (Respondent 5, Nurse, 6 years)

THEME 4: Training and Staff Competency

Training and staff competency were consistently cited as essential factors influencing the effective use of blood loss measurement tools in detecting postpartum hemorrhage (PPH). Many respondents emphasized that ongoing training and retraining of healthcare workers were crucial for ensuring proper usage of tools like suction machines, calibrated jars, and gauze. However, it was also noted that in many facilities, staff lacked adequate training or up-to-date knowledge about the latest techniques and tools for measuring blood loss. In some cases, lack of training led to inconsistent or incorrect application of available tools, which could ultimately affect patient outcomes.

Subtheme 4.1: Importance of Regular Training and Retraining

A significant number of respondents stressed the importance of regular training and retraining to ensure that healthcare workers were aware of new techniques, tools, and best practices for managing PPH. Regular training was seen as a way to improve skills, update knowledge, and ensure that healthcare workers were competent in using available blood loss measurement

tools. Respondents also noted that training would help eliminate outdated practices and equip staff with the confidence and competence to handle PPH more effectively.

"We need regular training on how to properly use the available tools. It's essential that we stay updated on new techniques, especially when we face cases like PPH." (Respondent 4, Nurse, 7 years)

"Training is key to making sure that everyone knows how to use the tools effectively. Without training, we risk using them incorrectly, which could lead to mismanagement." (Respondent 2, Midwife, 6 years)

"If there is no ongoing training, we fall behind. New tools and techniques come up, and we need to be taught how to use them correctly." (Respondent 3, Nurse, 5 years)

"We don't get retraining regularly, but it's crucial. I believe with better training, we could manage PPH more effectively with the tools we have." (Respondent 9, Nurse, 6 years)

"We have to have regular refresher courses. Sometimes we forget how to use the tools properly, and these courses would keep us sharp." (Respondent 8, Nurse-Midwife, 5 years)

"Training should be a continuous process. Without it, we might end up misusing tools or relying too much on visual estimation, which isn't ideal." (Respondent 5, Nurse, 12 years)

"The lack of up-to-date training leads to incorrect use of the tools, and that's a risk. We need training every year to stay on top of best practices." (Respondent 6, Midwife, 7 years)

"We don't have regular training sessions, and it shows. The nurses in our department aren't as proficient with the tools because of it." (Respondent 7, Nurse, 8 years)

Subtheme 4.2: Inadequate or Irregular Training Opportunities

While some respondents emphasized the importance of training, others noted that training opportunities were few or irregular, which impacted the overall competency of healthcare workers. Lack of funds, staffing issues, and limited access to workshops or seminars were frequently cited as barriers to proper and consistent training. In some instances, trainings were conducted, but they were sporadic and not regularly scheduled, leading to knowledge gaps and inconsistent practices in the use of blood loss measurement tools.

"We don't have enough opportunities for training. When we do have them, it's usually only once in a while, and sometimes the sessions are not thorough enough." (Respondent 4, Nurse, 6 years)

"In our hospital, training happens rarely, and when it does, it's usually not focused on the practical use of tools like suction machines or gauze." (Respondent 10, Midwife, 7 years)

"We need more regular workshops. The ones we get are too few and don't cover everything, so we miss out on some techniques that could help us." (Respondent 6, Nurse, 8 years)

"The training here is not regular, so we sometimes fall behind. When new tools or techniques are introduced, there isn't enough time for staff to learn them properly." (Respondent 5, Midwife, 6 years)

"In many cases, we learn on the job, but there should be structured training that's frequent enough to keep us updated." (Respondent 8, Nurse, 6 years)

"Training is more about theory than practice. We need more practical training to know exactly how to apply these tools in real situations." (Respondent 3, Nurse, 7 years)

"The lack of regular training means that people sometimes forget how to use the tools, and that can lead to inaccurate measurements of blood loss." (Respondent 7, Midwife, 5 years)

"There are some training sessions, but they don't happen often enough. When they do, they're usually not detailed enough to cover everything we need to know." (Respondent 2, Nurse, 6 years)

Subtheme 4.3: Mentorship and On-the-Job Training

Some respondents noted the value of mentorship and on-the-job training in the use of blood loss measurement tools. Experienced staff often provided guidance to newer or less experienced colleagues, but this system was not formalized or consistent. While this informal training could be effective, mentorship was often limited by the availability of senior staff or their willingness to invest time in mentoring others. Respondents suggested that a structured mentorship program could be beneficial for improving competency and ensuring that less experienced staff were properly trained in the use of tools.

"Mentorship is very helpful. As a junior nurse, I learn a lot from the senior nurses, especially when it comes to using the tools correctly." (Respondent 6, Nurse, 3 years)

"I had a mentor who showed me how to use the suction machine during my early years here. It's good to have someone guide you through the process." (Respondent 7, Nurse-Midwife, 6 years)

"We don't have a formal mentorship program, but the experienced nurses and doctors are always there to guide us when we need help." (Respondent 2, Nurse, 8 years)

"Mentorship on the job is useful, especially when you're first starting out. I was lucky to have senior staff who showed me how to handle the equipment correctly." (Respondent 5, Nurse, 6 years)

"A mentorship program would definitely help. Sometimes we learn from each other, but it's not as structured as it should be." (Respondent 9, Midwife, 4 years)

"Mentorship is great, but we don't have enough time or staff to make it more consistent." (Respondent 8, Nurse, 7 years)

"I've learned a lot from my colleagues, but it would be more efficient if mentorship was a regular part of the training program." (Respondent 6, Nurse, 9 years)

"Some people don't get enough mentorship, and that can lead to mistakes with the equipment. If it was more structured, it would help a lot." (Respondent 7, Nurse, 5 years)

4.3 The facilitators of the use of blood loss measurement tools in postpartum hemorhhage detection among health workers in Adeoyo Maternity Teaching Hospital Yemetu, Ibadan

Table 4.3: Themes and Sub-themes on Facilitators of the Use of Blood Loss Measurement Tools

Theme	Sub-theme
1. Support from Healthcare Providers	1.1 Role of doctors and midwives in encouraging use of tools 1.2 Collaboration and teamwork in tool utilization 1.3 Encouragement from supervisors and senior staff
2. Institutional Support and Resources	2.1 Availability of required tools and equipment 2.2 Financial resources for purchasing tools 2.3 Availability of adequate supplies (gauze, drapes, etc.)
3. Training and Professional Development	3.1 Continuous education and capacity building 3.2 Access to workshops, seminars, and training sessions 3.3 On-the-job training and practical application
4. Mentorship and Peer Learning	4.1 Learning from experienced colleagues 4.2 Importance of hands-on practice and guidance 4.3 Peer discussions and sharing of best practices
5. Positive Patient Outcomes	5.1 Improved patient care as motivation for using tools 5.2 Success stories and examples driving tool use
6. Managerial and Administrative Support	6.1 Support from hospital management in tool implementation 6.2 Clear communication of tool benefits from hospital leadership

HEME 1: Support from Healthcare Providers

Support from healthcare providers plays a crucial role in ensuring the effective use of blood loss measurement tools in postpartum hemorrhage (PPH) detection. Many participants highlighted that the active involvement of doctors, midwives, and senior staff was essential in motivating junior healthcare workers to properly utilize these tools. Encouragement and supervision from experienced professionals fostered a culture of accountability, teamwork, and effective tool usage. The presence of supportive leaders was particularly important in

ensuring that the necessary tools and practices were adhered to, even in challenging environments.

Subtheme 1.1: Role of Doctors and Midwives in Encouraging Use of Tools

Respondents consistently mentioned the significant role that doctors and midwives played in encouraging the use of blood loss measurement tools. Their leadership and supervisory role not only ensured that tools like suction machines, calibrated jars, and gauze were used correctly, but also helped create an environment in which junior staff felt confident in utilizing these tools. Doctors, especially obstetricians, and midwives often took the lead in ensuring that these tools were employed promptly and accurately to manage PPH cases.

"The doctors, especially the obstetricians, always encourage us to use the tools properly. They emphasize their importance during every PPH case, making sure we don't rely on estimation alone." (Respondent 4, Midwife, 7 years)

"The midwives are very instrumental in ensuring that blood loss tools are used correctly. They demonstrate their use during rounds, which makes a huge difference." (Respondent 5, Nurse, 6 years)

"The doctor always checks if we're using the right tools for PPH. If we miss something, they're quick to point it out, ensuring we follow the right procedure." (Respondent 6, Nurse, 9 years)

"I always feel more confident when the doctor is around. They guide us, and I know I'm doing it right because they watch over us and give us feedback." (Respondent 7, Nurse-Midwife, 5 years)

"Midwives play a key role. They don't just supervise us—they also guide us on what to do if a patient starts showing signs of PPH." (Respondent 8, Nurse, 4 years)

"In our hospital, the doctors are responsible for making sure all protocols, including the use of blood loss measurement tools, are strictly followed." (Respondent 3, Midwife, 6 years)

Subtheme 1.2: Collaboration and Teamwork in Tool Utilization

Collaboration and teamwork were cited as essential in ensuring the proper use of blood loss measurement tools. Many respondents emphasized the need for a multi-disciplinary approach,

where nurses, midwives, doctors, and anesthetists worked together to manage postpartum hemorrhage. Effective communication and coordination were seen as key factors that facilitated the proper use of tools during delivery and emergency situations. When all members of the healthcare team were on the same page regarding tool usage, the accuracy and efficiency of PPH management improved.

"Working with the team is important. When we all communicate about the patient's condition, including the use of blood loss tools, we can address PPH more effectively." (Respondent 4, Nurse, 6 years)

"During PPH cases, it's crucial that everyone is on the same page. Doctors, nurses, and midwives all have to be involved in using the tools correctly and promptly." (Respondent 9, Midwife, 8 years)

"Teamwork is key in managing PPH. We all rely on each other—whether it's the midwife assisting with gauze or the doctor monitoring blood loss." (Respondent 2, Nurse, 7 years)

"When a midwife and a nurse are working together, it's easier to ensure that the blood loss tools are used correctly. We coordinate well to get the job done." (Respondent 5, Nurse, 5 years)

"Collaboration with other team members helps us learn faster about using these tools, and we're all more confident in doing it right." (Respondent 6, Nurse-Midwife, 4 years)

"As part of the team, I'm constantly learning from the more experienced doctors and midwives. They make sure we're always using the tools as per the guidelines." (Respondent 8, Nurse, 6 years)

Subtheme 1.3: Encouragement from Supervisors and Senior Staff

Supervisors and senior healthcare workers were noted for providing critical encouragement and guidance in the use of blood loss measurement tools. Many participants expressed that having experienced staff who were willing to take the time to mentor and encourage junior staff helped foster an atmosphere where accuracy and accountability were prioritized. The feedback from supervisors allowed for continuous improvement in tool usage and PPH management.

"The senior nurses and doctors provide constant feedback, which helps us improve how we use the tools. They make sure we're doing it right and encourage us when we get it correct." (Respondent 7, Nurse-Midwife, 5 years)

"Our supervisor is very thorough about how we use the blood loss tools. They encourage us to make sure every measurement is accurate." (Respondent 9, Nurse, 8 years)

"It's important to have supervisors who are hands-on and ready to offer help when needed. They ensure we use the tools properly, and that gives us more confidence." (Respondent 6, Nurse, 9 years)

"Having senior staff who show us how to use the tools is reassuring. Their encouragement keeps us motivated to continue learning and using the tools correctly." (Respondent 4, Nurse, 6 years)

"When supervisors take the time to explain and show us how to use the blood loss measurement tools, we feel more competent in handling PPH cases." (Respondent 5, Nurse, 5 years)

"Sometimes, I feel unsure about whether I've used the tools properly, but senior staff always guide me and make sure I'm on the right track." (Respondent 3, Nurse-Midwife, 7 years)

THEME 2: Institutional Support and Resources

Institutional support and resources are crucial factors in enabling healthcare workers to use blood loss measurement tools effectively in postpartum hemorrhage (PPH) detection. The availability of tools, financial resources, and organizational policies that support the use of these tools ensure that healthcare providers can perform their duties effectively. This theme highlights the importance of having access to the right tools, equipment, and financial resources within the healthcare system. The support from the institution is essential not only for ensuring that the tools are available but also for maintaining a continuous supply and ensuring accessibility in emergency situations.

Subtheme 2.1: Availability of Required Tools and Equipment

Respondents frequently mentioned that having the right tools available in the healthcare facility was a fundamental requirement for effective PPH management. Tools such as calibrated jars, suction machines, gauze, and drapes were highlighted as essential for accurate blood loss measurement. However, some respondents noted that even when tools were available, there was still the issue of insufficient supply, which sometimes led to delays in PPH detection and intervention. The availability of these tools significantly improved healthcare workers' ability to make quick and accurate measurements of blood loss in postpartum hemorrhage cases.

"In this facility, we have the tools we need like calibrated jars and suction machines. But sometimes, we lack enough of them, especially when multiple PPH cases arise simultaneously." (Respondent 3, Nurse-Midwife, 6 years)

"The tools are available, but there are not enough suction machines. In some cases, we have to wait for the machine to be free before using it, which can delay care." (Respondent 5, Nurse, 4 years)

"We have the basic tools like gauze and calibrated jars, but the number of them isn't always enough to meet the demand during emergencies." (Respondent 8, Midwife, 7 years)

"The facility provides the necessary equipment, but sometimes they run out, especially gauze and drapes, which are essential for accurate blood loss measurement." (Respondent 4, Nurse, 6 years)

"The hospital ensures we have the tools needed, but there's often a shortage of specific items like the drapes, which makes it hard to get precise measurements." (Respondent 9, Midwife, 8 years)

Subtheme 2.2: Financial Resources for Purchasing Tools

Financial resources were also highlighted as a significant factor in ensuring that the required tools and equipment were available. Respondents pointed out that the funding constraints often hindered the ability of healthcare facilities to purchase enough blood loss measurement tools. While hospital management tried to provide the necessary tools, budget limitations and financial challenges in the healthcare system meant that some essential tools were unavailable

or in short supply. In some cases, healthcare workers were required to source their own tools, or patients were expected to buy some of the necessary equipment.

"We always face financial challenges. The hospital doesn't always have enough funds to purchase more tools, which affects our ability to handle multiple PPH cases effectively." (Respondent 6, Nurse, 9 years)

"Sometimes we have to ask patients to bring their own gauze or drapes because the hospital can't always afford to buy enough." (Respondent 7, Nurse-Midwife, 5 years)

"The hospital has some financial constraints that prevent them from buying all the equipment we need. This often leads to shortages of crucial tools like suction machines." (Respondent 8, Nurse, 4 years)

"There's not enough funding for the hospital to purchase enough drapes, and that impacts our ability to measure blood loss accurately during PPH." (Respondent 2, Midwife, 6 years)

"Financial constraints make it difficult to replace worn-out equipment, and that affects how well we can perform blood loss measurements." (Respondent 4, Nurse, 6 years)

Subtheme 2.3: Availability of Adequate Supplies (Gauze, Drapes, etc.)

The availability of adequate supplies was mentioned as a key factor for the effective use of blood loss measurement tools. Respondents reported that while some tools, like gauze and suction machines, were available, other necessary supplies like drapes or calibrated jars were often insufficient or not replaced on time. This led to delays in providing accurate measurements and, in some cases, reliance on visual estimation rather than actual measurement.

"Gauze and suction machines are readily available, but sometimes drapes and calibrated jars are in limited supply. We often have to rely on visual estimation because the equipment is not enough." (Respondent 7, Nurse, 5 years)

"While we have enough gauze, drapes are in short supply. It's frustrating because accurate measurement depends on them." (Respondent 5, Nurse, 6 years)

"We have the basic tools, but sometimes the lack of adequate drapes and calibrated jars means we can't get accurate measurements for PPH." (Respondent 3, Midwife, 6 years)

"The facility does its best to provide what we need, but there are always gaps. The lack of drapes means we sometimes rely on less accurate methods." (Respondent 9, Nurse, 4 years)

"The hospital has been able to provide gauze and suction machines, but we're often left without drapes when they're most needed." (Respondent 8, Nurse, 5 years)

THEME 3: Training and Professional Development

Training and professional development play a critical role in ensuring the effective use of blood loss measurement tools in postpartum hemorrhage (PPH) detection. Many respondents emphasized the importance of ongoing training to ensure that healthcare workers are proficient and competent in using these tools. Continuous education, including workshops, seminars, and on-the-job training, was identified as key to improving knowledge and skills among healthcare providers. In addition to improving their technical skills, training also helped boost confidence and reduced errors in blood loss estimation. However, some respondents also expressed concerns about the lack of adequate training or the infrequency of training programs, especially in the context of new innovations in PPH management. The gaps in training sometimes resulted in reliance on visual estimation rather than accurate tool-based measurement.

Subtheme 3.1: Continuous Education and Capacity Building

Continuous education and capacity building were highlighted as essential for ensuring that healthcare workers are well-equipped to use blood loss measurement tools effectively. Respondents emphasized the need for regular refresher courses and professional development programs to keep healthcare workers up to date with the latest practices and technological advancements. Many respondents acknowledged that knowledge of current tools and methods was crucial for accurately assessing postpartum hemorrhage.

"We have periodic workshops that help us stay up to date with the latest tools and techniques for measuring blood loss." (Respondent 5, Nurse-Midwife, 7 years)

"Continuous education is vital. Whenever there's a new method or tool, we have to attend workshops to learn how to use them effectively." (Respondent 6, Midwife, 8 years)

"I feel more confident using the tools now because of the training I've received. It's important that we keep learning, especially as technology advances." (Respondent 4, Nurse, 6 years)

"Training programs help reinforce the importance of accurate measurement, and without those, we would be relying on guesswork more often." (Respondent 8, Nurse, 5 years)

"We are regularly taught how to properly use the tools, and I think this keeps us sharp and reduces errors when managing PPH." (Respondent 9, Nurse-Midwife, 6 years)

"The workshops make a huge difference. Without them, many of us wouldn't know the proper way to measure blood loss during PPH." (Respondent 7, Nurse, 5 years)

Subtheme 3.2: Access to Workshops, Seminars, and Training Sessions

Access to workshops, seminars, and other professional development opportunities was frequently mentioned as a facilitator of tool usage. Formal training sessions offered through the healthcare facility or external professional bodies helped healthcare workers stay informed about new techniques, tools, and methods for PPH management. However, some respondents noted that the availability of such sessions could be limited, especially in under-resourced facilities or in regions where continuing education programs were not as easily accessible.

> *"We get invited to attend seminars, which are helpful, but sometimes we don't have enough time to attend them, especially with our busy schedules."* (Respondent 6, Nurse, 7 years)
>
> *"There are training sessions organized by the hospital, but not all staff can attend due to shift schedules or the lack of sufficient resources to support attendance."* (Respondent 4, Midwife, 6 years)
>
> *"We often receive information from seminars and workshops, but the funding for travel or attendance is sometimes an issue."* (Respondent 5, Nurse, 5 years)
>
> *"The workshops we attend give us the confidence to use the tools more accurately. It's an essential part of our professional growth."* (Respondent 8, Nurse-Midwife, 7 years)
>
> *"The hospital facilitates access to some training, but sometimes there's just not enough time or support for everyone to participate."* (Respondent 9, Nurse, 6 years)

"We do have access to workshops, but I think we need more frequent ones, especially on the new tools being introduced." (Respondent 3, Nurse, 8 years)

Subtheme 3.3: On-the-Job Training and Practical Application

On-the-job training and hands-on practice were frequently mentioned as essential for mastering the use of blood loss measurement tools. Respondents emphasized the importance of practical experience, where they could apply their knowledge in real-life situations under the guidance of experienced staff. Many reported that this type of training helped them develop confidence and competency in measuring blood loss accurately, especially in high-pressure situations like PPH cases.

"The real learning happens when we're actively using the tools during deliveries. Supervisors guide us, and we get hands-on experience." (Respondent 7, Nurse, 6 years)

"On-the-job training has been the most effective for me. The more I use the tools during real cases, the more confident I become." (Respondent 9, Nurse-Midwife, 5 years)

"Nothing beats learning on the job. When we actually use the blood loss tools during a PPH case, it's easier to understand their importance." (Respondent 4, Nurse, 6 years)

"Supervisors guide us during deliveries, showing us how to use the tools in real-time. It's this hands-on training that makes all the difference." (Respondent 6, Nurse, 5 years)

"We practice with the tools during our shifts. This way, we learn how to use them in different scenarios, not just in theory." (Respondent 8, Nurse, 7 years)

"Having experienced nurses and midwives teach us while we're on the job really boosts our skills. It's how we learn best." (Respondent 3, Nurse, 6 years)

THEME 4: Mentorship and Peer Learning

Mentorship and peer learning emerged as essential components in the use of blood loss measurement tools for postpartum hemorrhage (PPH) detection. Many respondents highlighted the value of learning from experienced colleagues, hands-on practice, and peer

discussions in improving their competence and confidence in using the available tools. These mechanisms of mentorship and collaboration were often seen as key enablers for healthcare workers to refine their skills in a real-world setting. In addition to formal training, these informal learning methods played a crucial role in practical application and knowledge-sharing. However, some respondents also pointed out that these opportunities could be limited by time constraints, staff shortages, and lack of structured mentorship programs in certain facilities.

Subtheme 4.1: Learning from Experienced Colleagues

Learning from experienced colleagues was frequently cited as an important facilitator in improving blood loss measurement skills. Many healthcare workers, especially newer staff, relied on the guidance and expertise of more experienced colleagues to better understand the technicalities of using blood loss measurement tools. These mentors provided valuable advice on tool handling, methodology, and how to navigate challenges that arose in PPH cases. The learning-by-doing approach allowed healthcare workers to gain confidence in accurately measuring blood loss and understanding its clinical significance.

> *"I learned most of what I know about blood loss measurement from more experienced nurses. They show me how to use the tools properly during our shifts."* (Respondent 4, Nurse, 7 years)
>
> *"Our senior nurses are always there to guide us through the process of measuring blood loss. They help us understand when to use which tool and how to interpret results."* (Respondent 6, Midwife, 6 years)
>
> *"I've learned a lot from my colleagues, especially when they demonstrate how to use the gauze or suction machine during real cases. It's been helpful for me."* (Respondent 9, Nurse, 5 years)
>
> *"The senior nurses have taught me how to estimate blood loss when the calibrated jars are not available. Their experience has been invaluable."* (Respondent 3, Nurse, 6 years)

"I didn't understand the use of calibrated jars initially, but my mentor explained it in a way that made sense, and I got to practice under supervision." (Respondent 5, Nurse, 7 years)

"Most of the guidance I've received comes from the more experienced staff members. They walk us through difficult cases, explaining how to measure and what to look for." (Respondent 7, Nurse-Midwife, 5 years)

Subtheme 4.2: Importance of Hands-On Practice and Guidance

The importance of hands-on practice and guidance from mentors was emphasized as crucial for developing proficiency in using blood loss measurement tools. Respondents stressed that the best way to learn how to measure blood loss accurately was through practical experience, where they could apply what they had learned in training sessions under the supervision of senior staff. This direct involvement in real-life scenarios helped to solidify their understanding and gave them the opportunity to correct mistakes in a supportive environment.

"You can read all you want about blood loss measurement, but the real learning happens when you're actually doing it, with a mentor watching and guiding you." (Respondent 5, Nurse, 7 years)

"Hands-on practice is the best way to get better at using the tools. Our mentors let us practice under supervision so we can learn and make mistakes in a safe environment." (Respondent 9, Midwife, 6 years)

"Mentors provide hands-on guidance. They show us exactly how to handle the tools during real patient care, which builds our confidence." (Respondent 3, Nurse, 6 years)

"Being shown how to use the calibrated jar in an actual PPH case made me feel more confident than just reading about it." (Respondent 7, Nurse, 5 years)

"I have learned the most when I've been able to practice the tools in real-life situations, with my mentor providing feedback and corrections." (Respondent 8, Midwife, 5 years)

"It was the hands-on experience that really helped me. My mentor guided me through the steps, and I practiced until I was comfortable." (Respondent 6, Nurse, 7 years)

Subtheme 4.3: Peer Discussions and Sharing of Best Practices

Peer discussions and the sharing of best practices were also crucial in improving the use of blood loss measurement tools. Respondents mentioned that informal conversations with

colleagues allowed them to exchange experiences and discuss challenges they encountered when using the tools. These peer-to-peer interactions often led to the discovery of better techniques, troubleshooting methods, and the sharing of successful strategies. This collaborative learning was especially important in cases where mentorship opportunities were limited, and it allowed healthcare workers to fill knowledge gaps by discussing different approaches and solutions.

"We often have informal discussions in the break room where we share tips on how to handle blood loss situations more effectively. It's helpful to hear what works for others." (Respondent 4, Nurse, 6 years)

"Peer discussions have been great for learning about different methods for measuring blood loss. Sometimes, one person might have a trick or technique that really works." (Respondent 6, Nurse, 7 years)

"I learned a lot from talking to my colleagues about how they handle PPH. We share our experiences and learn from each other." (Respondent 9, Midwife, 6 years)

"Whenever we face difficulties with the tools, we always talk to each other to find the best solutions. Peer discussions are crucial for improving our practice." (Respondent 3, Nurse, 5 years)

"It's good to talk with colleagues about how we can improve our measurements. We all have our own approaches, but by sharing ideas, we help each other do better." (Respondent 5, Nurse, 6 years)

"Our team often discusses how we measure blood loss, and we share what's working for us. Sometimes it's the little things that make the biggest difference." (Respondent 7, Midwife, 6 years)

THEME 5: Positive Patient Outcomes

Positive patient outcomes were a significant motivator for healthcare workers to utilize blood loss measurement tools in postpartum hemorrhage (PPH) detection. Healthcare workers highlighted that the use of accurate tools led to improved patient care, particularly in terms of early intervention, prevention of maternal morbidity, and enhanced recovery. Healthcare

workers who saw positive results in their patients were more motivated to adopt and use the tools consistently. The success stories shared by colleagues about how the tools contributed to better patient outcomes further strengthened their commitment to accurate blood loss measurement.

Subtheme 5.1: Improved Patient Care as Motivation for Using Tools

Healthcare workers emphasized that improved patient care was the primary reason for using blood loss measurement tools. Accurate blood loss measurements allowed them to intervene early, leading to better outcomes for patients. When they saw that these tools helped in early detection and timely interventions, preventing severe complications, it motivated them to keep using the tools. Respondents agreed that the tools facilitated better patient care and helped them deliver safer care.

"Using these tools is key to saving lives. Early detection means we can prevent a lot of complications. I've seen it myself, and that's why I use them every time." (Respondent 1, Nurse, 10 years)

"We see the difference in patient care when we use the tools correctly. It helps us act fast and prevent severe hemorrhage." (Respondent 2, Midwife, 7 years)

"It's motivating when we can see the improvement in patients. With the proper measurement tools, we catch issues early and prevent fatalities." (Respondent 6, Nurse, 6 years)

"Using the tools ensures we act fast. When we measure blood loss properly, we can intervene earlier, which ultimately saves lives." (Respondent 9, Nurse, 5 years)

"Patient care improves drastically when you can accurately measure blood loss. I've seen it myself when the intervention was quick, and the patient's health improved." (Respondent 11, Midwife, 6 years)

"I have seen it in real life—the difference that using the right tools can make. It motivates me to use them every time." (Respondent 15, Nurse, 5 years)

"The use of tools directly impacts patient recovery. When we use them properly, the chances of complications decrease." (Respondent 12, Nurse, 8 years)

"I remember a case where we caught severe bleeding early, thanks to the blood loss measurement tools. It saved that patient's life." (Respondent 18, Nurse, 7 years)

Subtheme 5.2: Success Stories and Examples Driving Tool Use

Success stories shared by colleagues about the positive outcomes of using blood loss measurement tools were powerful motivators for others to adopt these tools. Hearing about specific cases where the tools prevented maternal morbidity or improved patient recovery encouraged healthcare workers to implement them regularly. These real-life examples of successful interventions showcased the value of using accurate blood loss measurements and reinforced the tools' role in enhancing patient care.

"A colleague once shared how using the suctioning tool helped save a woman's life during a cesarean. That really pushed me to use the tools more often." (Respondent 5, Midwife, 6 years)

"I heard a nurse share a story about how using the tools helped save a woman from severe blood loss. That kind of success story made me more committed to using the tools." (Respondent 6, Nurse, 7 years)

"One of my colleagues had a patient with severe hemorrhage, but because the tools were used, we were able to control the bleeding. Hearing that success made me more confident in using them." (Respondent 10, Nurse, 5 years)

"A senior midwife shared an amazing story where using the tools early on helped a patient avoid a major complication. That was a big eye-opener for me, and I started using the tools more regularly." (Respondent 7, Nurse, 6 years)

"The success stories from my colleagues really pushed me to adopt the use of these tools. I saw how lives were saved and wanted to be part of that change." (Respondent 3, Nurse, 6 years)

"When my colleagues shared success stories about how using these tools helped in critical moments, I was motivated to follow suit. I saw how it made a difference in patient outcomes." (Respondent 8, Nurse, 7 years)

"Success stories from colleagues made me realize how important these tools are. I've seen firsthand how using them leads to better outcomes." (Respondent 14, Midwife, 8 years)

"Every time I hear about a colleague's success with these tools, I'm reminded of their importance. It's motivating to know that they really do save lives." (Respondent 20, Nurse, 5 years)

THEME 6: Managerial and Administrative Support

Managerial and administrative support was recognized as a vital element for the successful implementation and use of blood loss measurement tools in the detection of postpartum hemorrhage (PPH). Respondents pointed out that active involvement from hospital management, including the provision of necessary resources, equipment, and training, significantly impacted the consistent use of the tools. Clear communication from leadership regarding the importance of the tools further supported their adoption. When healthcare workers felt that management and leadership were committed to improving patient care through effective tools, they were more likely to prioritize and incorporate these practices into their daily routines.

Subtheme 6.1: Support from Hospital Management in Tool Implementation

Support from hospital management was essential in providing the necessary resources for the successful implementation of blood loss measurement tools. Management's efforts to ensure that essential tools, such as suction machines, calibrated jars, and blood loss measurement drapes, were available helped create a more structured approach to accurate blood loss measurement. When hospital management actively supported these initiatives, healthcare workers were more motivated to use the tools, recognizing that management was committed to improving patient care and ensuring the availability of adequate resources.

"Hospital management played a big role in making sure we had the tools we needed. Without their support, we wouldn't have had the resources to implement these practices." (Respondent 10, Nurse, 7 years)

"The support from the administration was key in making sure we had the proper equipment, like calibrated jars and suction machines. This made it much easier for us to use the tools effectively." (Respondent 5, Nurse, 6 years)

"When the management showed commitment by providing the necessary equipment, it made it easier for us to follow best practices. Their support was vital in implementing these tools." (Respondent 2, Midwife, 6 years)

"Our hospital leadership ensured that the tools were available, which made it easier for us to integrate them into our practice." (Respondent 16, Nurse, 5 years)

"Hospital management's active support made a difference. They made sure the tools we needed were there, and that encouraged us to use them more often." (Respondent 12, Midwife, 8 years)

"Without management's backing, we wouldn't have had the equipment. Their support helped in getting everything we needed, from drapes to suction devices." (Respondent 14, Nurse, 6 years)

"The fact that the management ensured we had the right tools available made it easier for us to implement the correct procedures." (Respondent 8, Nurse, 7 years)

"Management helped in organizing the purchase of proper equipment, which really enhanced our ability to measure blood loss accurately." (Respondent 18, Midwife, 7 years)

Subtheme 6.2: Clear Communication of Tool Benefits from Hospital Leadership

The communication of the tools' benefits from hospital leadership played a significant role in motivating healthcare workers to use blood loss measurement tools regularly. When hospital leaders clearly communicated the importance of these tools in improving patient care and preventing maternal complications, it encouraged healthcare workers to prioritize their use. Leadership's advocacy made it clear that using the tools wasn't just a procedural requirement but a vital practice for patient safety and better health outcomes. This clarity from leadership reinforced the idea that using the tools aligned with the hospital's commitment to excellence in care.

"The leadership's communication about the importance of these tools really helped us understand how they contributed to better patient care. It made me more committed to using them." (Respondent 4, Nurse, 6 years)

"Our leadership communicated clearly why the tools were necessary—explaining how they improve patient outcomes. It helped everyone understand that these tools were essential to our work." (Respondent 3, Midwife, 5 years)

"Hospital leadership regularly communicated how important these tools were in improving care, which motivated me to adopt them in my practice." (Respondent 13, Nurse, 7 years)

"The leadership made sure we understood that using the tools would help us provide better care. That message was clear from day one, and it made me more motivated to use them." (Respondent 9, Midwife, 6 years)

"Leadership was vocal about the importance of blood loss measurement. They made it clear that these tools were essential for providing high-quality care to our patients." (Respondent 7, Nurse, 5 years)

"The administration has always made sure to remind us of the benefits of using these tools, and that kept us focused on patient safety." (Respondent 15, Nurse, 6 years)

"Hospital leadership made it clear that these tools were not optional—they were critical to the care we provide. That clear communication reinforced the need for us to use them." (Respondent 6, Nurse, 5 years)

"By emphasizing the benefits of these tools for patient safety, leadership ensured that we understood the real-world value of using them." (Respondent 17, Midwife, 7 years)

4.5 The barriers that hinder blood loss measurement tools in postpartum hemorrhage detection among health workers in Adeoyo Maternity Teaching Hospital Yemetu, Ibadan

Table4.6: Themes and Sub-themes on Barriers to the Use of Blood Loss Measurement Tools

Theme	Sub-theme
1. Inadequate Resources and Equipment	1.1 Shortage of key tools (suction machines, calibrated jars, blood drapes) 1.2 Limited availability of functional equipment
2. Financial Constraints	2.1 Insufficient funding for procurement of measurement tools 2.2 Out-of-pocket expenses for patients impacting tool use
3. Lack of Training and Knowledge	3.1 Insufficient training opportunities for healthcare workers 3.2 Limited knowledge on the proper use of measurement tools
4. Workload and Staffing Issues	4.1 High patient-to-staff ratio impacting effective tool use 4.2 Insufficient time to accurately measure blood loss
5. Institutional and Systemic Challenges	5.1 Poor management support and lack of policy enforcement 5.2 Lack of integration of tool use into standard protocols

THEME 1: Inadequate Resources and Equipment

Inadequate resources and equipment emerged as one of the primary barriers to the use of blood loss measurement tools in postpartum hemorrhage (PPH) detection. Healthcare workers noted that the lack of essential tools such as suction machines, calibrated jars, and blood drapes significantly hindered the accurate measurement of blood loss. Furthermore, the limited availability of functional equipment meant that healthcare workers often had to resort to improvised methods or visual estimation, which undermined the effectiveness of their practice and compromised patient care.

Subtheme 1.1: Shortage of Key Tools (Suction Machines, Calibrated Jars, Blood Drapes)

One of the most frequently reported issues by healthcare workers was the shortage of key measurement tools such as suction machines, calibrated jars, and blood drapes. Many healthcare workers expressed frustration with the unavailability of these tools, which are critical for accurate blood loss measurement. As a result, they were often forced to improvise, relying on visual estimation or using whatever limited resources were available to them. This shortage hindered their ability to deliver the best care and led to uncertainty in the management of postpartum hemorrhage.

"We don't always have enough suction machines. Most of the time, we have to make do with one or two, and we have to share between wards." (Respondent 6, Nurse, 7 years)

"The suction machines are always in short supply here. Sometimes, we can't use them at all because they're either broken or out of service." (Respondent 3, Midwife, 6 years)

"We rely on visual estimation most of the time because we don't have enough calibrated jars to accurately measure blood loss." (Respondent 12, Nurse, 5 years)

"I've only seen a blood drape once or twice in my career. It's not something we usually have access to in this hospital." (Respondent 9, Nurse, 6 years)

"The gauze and pads are what we mostly use, and even then, we have to guess how much blood is soaked because we lack accurate tools like drapes." (Respondent 14, Midwife, 7 years)

"If we had more suction machines, it would be much easier to measure blood loss accurately, but we don't have enough resources." (Respondent 5, Nurse, 6 years)

"We just don't have the tools to do it properly. A suction machine here and there, and that's it. The rest we do with visual estimation." (Respondent 8, Nurse, 5 years)

"Lack of drapes is a big problem. Without them, we have to rely on makeshift methods, which don't provide reliable results." (Respondent 7, Nurse, 6 years)

Subtheme 1.2: Limited Availability of Functional Equipment

Even when the necessary equipment was available, it was often not functional or was in such poor condition that it could not be relied upon for accurate measurements. Respondents highlighted that even when equipment such as suction machines and calibrated jars were provided, they were not always in working order or were frequently out of service, causing delays and increasing the likelihood of errors in measurement. This instability in equipment availability added a layer of uncertainty to the healthcare workers' ability to properly measure blood loss and intervene in a timely manner.

"We don't always have the tools in working condition. Some of the suction machines are broken, and even when they work, we have to share them between different wards." (Respondent 4, Nurse, 5 years)

"The calibrated jars we have sometimes don't even have clear markings anymore, making it hard to use them accurately. I've had to rely on old jars that are not as clear as they should be." (Respondent 10, Midwife, 7 years)

"Even when we have the tools, we can't always count on them. Sometimes the suction machine just stops working in the middle of a procedure." (Respondent 13, Nurse, 7 years)

"The suction devices are often unreliable. I've had to ask other departments to borrow equipment, but it delays the process and causes a lot of stress." (Respondent 6, Nurse, 6 years)

"The lack of functional equipment is a major problem. Even when we get the tools, we can't always use them because they are not well maintained." (Respondent 15, Midwife, 6 years)

"We have the equipment, but they don't work all the time. Sometimes we just have to make do with whatever is functional in the moment." (Respondent 2, Nurse, 5 years)

"We've had situations where the tools broke down in the middle of an emergency. It's frustrating because it compromises patient care." (Respondent 17, Nurse, 6 years)

THEME 2: Financial Constraints

Financial constraints emerged as another significant barrier to the proper utilization of blood loss measurement tools in the detection of postpartum hemorrhage (PPH). Healthcare workers noted that limited funding for necessary tools and resources made it difficult for hospitals to procure the required equipment. Furthermore, the cost of equipment and the financial burden on patients for out-of-pocket expenses for tools and supplies were additional challenges that hindered the effective use of blood loss measurement tools. These financial barriers not only impacted the availability of essential tools but also created additional stress for healthcare workers who were unable to provide optimal care due to limited resources.

Subtheme 2.1: Insufficient Funding for Procurement of Measurement Tools

Many respondents highlighted the lack of adequate funding for medical equipment as a critical barrier to the availability of blood loss measurement tools. Hospital management often faced financial difficulties, which impacted their ability to purchase new equipment or replace faulty tools. This lack of funding meant that essential tools like calibrated jars, suction machines, and blood drapes were either inadequate or unavailable. As a result, healthcare workers had to compensate with visual estimation or use improvised methods, which were far from ideal.

"We don't have enough money to buy the necessary equipment. Many times, we make do with what we have, which is not always the best." (Respondent 7, Nurse, 6 years)

"The hospital doesn't have enough funds to keep the equipment running. We have to wait until something breaks down before they look for money to fix it." (Respondent 11, Midwife, 8 years)

"The financial resources aren't enough to keep up with the demand for the equipment. We don't have the luxury of buying new tools regularly." (Respondent 2, Nurse, 5 years)

"We often face a lack of funding. It takes too long for management to approve the purchase of essential tools like calibrated jars." (Respondent 10, Midwife, 6 years)

"I've had to wait for months for certain equipment to be replaced because the hospital can't afford it. Meanwhile, we continue working with outdated tools." (Respondent 14, Nurse, 7 years)

"Funds for new equipment are a constant issue. Even when we know the tools are essential, we can't always get them due to the budget limitations." (Respondent 13, Midwife, 7 years)

"The hospital is underfunded, and this affects everything, including the tools we need to measure blood loss." (Respondent 9, Nurse, 6 years)

Subtheme 2.2: Out-of-pocket Expenses for Patients Impacting Tool Use

In addition to hospital financial constraints, respondents also pointed out that many patients were required to cover the cost of certain supplies themselves. This was particularly problematic for tools such as gauze, pads, and blood drapes, which were often not provided by the hospital and had to be purchased by the patients. The inability of many patients to afford these costs contributed to inaccurate blood loss measurements and sometimes led to delays in intervention when tools were unavailable. These financial burdens on patients exacerbated the challenges healthcare workers faced when trying to ensure accurate and timely measurement of blood loss.

"Many patients can't afford the required pads or gauze. So, we often have to rely on what they bring, which isn't always adequate." (Respondent 8, Nurse, 5 years)

"Patients struggle to pay for things like gauze and drapes, and sometimes the hospital doesn't cover these costs. This creates delays in care." (Respondent 15, Midwife, 6 years)

"Some patients have to buy their own gauze and pads. This out-of-pocket expense is a huge burden, especially for those who can't afford it." (Respondent 16, Nurse, 5 years)

"We can't use some of the tools because patients are required to pay for them upfront, and many can't afford it." (Respondent 14, Midwife, 7 years)

"I've had to use gauze that patients bring in themselves. Some of them can only afford cheap ones, which are not reliable." (Respondent 12, Nurse, 6 years)

"It's frustrating when patients can't afford the tools because it slows down our ability to do our job properly. We have to keep estimating instead of using the right equipment." (Respondent 6, Nurse, 7 years)

"If patients can't pay for the drapes or gauze, we just have to use the limited resources we have, which affects how accurately we can measure blood loss." (Respondent 18, Midwife, 7 years)

THEME 3: Lack of Training and Knowledge

Lack of training and knowledge emerged as a crucial barrier to the effective use of blood loss measurement tools in the detection of postpartum hemorrhage (PPH). Respondents frequently reported that insufficient training opportunities for healthcare workers, particularly in the proper use of measurement tools, contributed to ineffective practices. Many participants noted that inadequate knowledge about the correct protocols for measuring blood loss and use of measurement tools led to incorrect estimations and delayed interventions. Furthermore, a lack of continuous education and professional development opportunities for healthcare workers was cited as an obstacle to improving the quality of care.

Subtheme 3.1: Insufficient Training Opportunities for Healthcare Workers

One of the most frequently reported issues was the insufficient training opportunities for healthcare workers on the proper use of blood loss measurement tools. Many respondents noted that regular workshops, seminars, and refresher courses on postpartum hemorrhage management were not conducted frequently enough, leaving workers without the necessary skills to effectively use the available tools. Some respondents also mentioned that there was no structured training program to ensure that new staff were adequately prepared for using

measurement tools. This lack of training created a knowledge gap, which led to reliance on outdated practices and inconsistent use of tools.

"We don't get enough training on how to use the tools. I think we need regular workshops to update our skills." (Respondent 9, Nurse, 6 years)

"Training is sporadic. Sometimes, we go months without any training, and when there is, it's usually very basic." (Respondent 5, Midwife, 7 years)

"The hospital doesn't have a system for continuous training, so many of us don't feel confident in using the tools correctly." (Respondent 3, Nurse, 5 years)

"I have never received formal training on how to use calibrated jars or suction machines for measuring blood loss." (Respondent 6, Midwife, 6 years)

"The training on postpartum hemorrhage management is inconsistent, and it's usually just theoretical, not enough practical demonstrations." (Respondent 13, Nurse, 5 years)

"We need more hands-on training. Without it, it's hard to be confident about using these tools correctly." (Respondent 10, Midwife, 7 years)

"We don't have specific training on using the gauze or drapes for blood loss measurement. It's just assumed we know how to do it." (Respondent 14, Nurse, 5 years)

Subtheme 3.2: Limited Knowledge on the Proper Use of Measurement Tools

Respondents also highlighted that a lack of proper knowledge regarding the functions and correct use of blood loss measurement tools was a major barrier. This knowledge gap meant that healthcare workers often made incorrect estimates or relied on inconsistent methods. Some healthcare workers were not fully aware of the calibration process for certain tools, like suction machines and calibrated jars, and others lacked clarity on the measurement protocols for postpartum hemorrhage. This subtheme points to the importance of formal education and hands-on experience in ensuring that healthcare workers have the knowledge and confidence to use these tools effectively.

"I wasn't properly trained on the calibration of the suction machines. Sometimes, I don't even know if the readings are accurate." (Respondent 7, Nurse, 6 years)

"There is confusion about when to use the suction machines or when to use gauze. We haven't had any real training on the specifics of these tools." (Respondent 12, Midwife, 5 years)

"The knowledge about blood loss measurement is limited. We don't know how much blood is in a soaked pad or gauze because we haven't been trained properly on this." (Respondent 8, Nurse, 7 years)

"I've had to learn on the job because there's no structured training program. Sometimes, I just follow what others are doing." (Respondent 15, Nurse, 5 years)

"We rely on visual estimation most of the time because we were never taught to measure blood loss accurately using tools like the calibrated jar." (Respondent 6, Midwife, 6 years)

"I think part of the reason we struggle with using the tools correctly is because we weren't adequately trained on them during our induction." (Respondent 3, Nurse, 7 years)

"We haven't been taught the importance of using calibrated tools. We just do what we've always done, which isn't always accurate." (Respondent 11, Midwife, 6 years)

THEME 4: Workload and Staffing Issues

Healthcare workers consistently highlighted workload and staffing issues as major barriers to the effective use of blood loss measurement tools in postpartum hemorrhage (PPH) detection. These issues were primarily related to high patient-to-staff ratios and the insufficient time available to accurately measure blood loss. Respondents reported that the high workload led to stressful working conditions, which made it challenging to focus on precision when using measurement tools. As a result, many healthcare workers rushed the measurement process, and in some cases, visual estimation or substandard practices were used as substitutes for the accurate measurement of blood loss.

Subtheme 4.1: High Patient-to-Staff Ratio Impacting Effective Tool Use

The high patient-to-staff ratio was repeatedly cited as a major barrier to using blood loss measurement tools effectively. Respondents noted that, particularly in busy hospitals like Adeoyo Maternity Teaching Hospital, the number of patients often exceeded the available staff, which led to increased pressure on healthcare workers. The staffing shortages resulted

in inefficient monitoring and rushed assessments. Due to these constraints, healthcare workers reported that they were unable to provide adequate care to each patient, which directly impacted the accurate use of blood loss measurement tools.

"There are too many patients, and we only have a few nurses. We're always in a rush, and it's difficult to focus on measuring blood loss accurately." (Respondent 14, Nurse, 5 years)

"The high number of patients means that we have to prioritize urgent care, and tools for measuring blood loss sometimes take a backseat." (Respondent 16, Midwife, 6 years)

"Because we are understaffed, sometimes the task of measuring blood loss is delegated to whoever is available, and we don't always have the time to do it properly." (Respondent 10, Nurse, 7 years)

"It's difficult to focus on blood loss measurement when there are so many patients to attend to. We often just estimate to move on quickly." (Respondent 8, Nurse, 6 years)

"Our team is overwhelmed with patients. The nurse-to-patient ratio is so high, we can't give the proper attention to things like blood loss measurement." (Respondent 11, Midwife, 7 years)

Subtheme 4.2: Insufficient Time to Accurately Measure Blood Loss

The lack of time due to high workload was another significant barrier reported by healthcare workers. Many respondents mentioned that they were often pressured by time constraints and could not take the necessary time to measure blood loss accurately. Some nurses and midwives described how limited time prevented them from thoroughly checking all the factors involved in blood loss measurement, such as the calibration of equipment or ensuring that all tools were used correctly. As a result, some healthcare workers resorted to visual estimation instead of proper measurement.

"When we have a full ward, there's no time to measure blood loss properly. We just do what we can and move on." (Respondent 5, Nurse, 6 years)

"We barely have time to weigh the gauze properly. Most times, we just estimate based on how much we think has been soaked." (Respondent 12, Midwife, 6 years)

"Because of the time pressure, we don't always get to use the calibrated jars as accurately as we should. There's just no time." (Respondent 9, Nurse, 7 years)

"Time is always a challenge. We have a lot of patients to take care of, so we can't afford to spend too much time on measuring blood loss." (Respondent 7, Midwife, 5 years)

"I wish we had more time to do it right, but the rush is always there. Sometimes, it's just about getting it done." (Respondent 3, Nurse, 5 years)

THEME 5: Institutional and Systemic Challenges

Institutional and systemic challenges were also significant barriers to the effective use of blood loss measurement tools. Respondents identified a lack of management support and poor policy enforcement as issues hindering the integration of blood loss measurement into routine care. Many participants pointed out that hospital management did not prioritize the implementation and sustainability of blood loss measurement tools, and there was no clear policy or protocol guiding their use. Additionally, the lack of integration of blood loss measurement into the standard care protocols and the absence of routine audits contributed to inconsistent practices and the underuse of available tools.

Subtheme 5.1: Poor Management Support and Lack of Policy Enforcement

Respondents consistently reported that poor management support and the lack of policy enforcement were barriers to the effective use of blood loss measurement tools. Many healthcare workers expressed frustration that hospital leadership did not provide adequate resources or support for implementing and maintaining blood loss measurement tools. The absence of clear policies on how and when to use the tools also contributed to inconsistent practices. Respondents noted that management rarely enforced standard operating procedures or provided the necessary oversight to ensure that the tools were used effectively.

"The management doesn't emphasize the importance of measuring blood loss. There are no clear policies, so it's often up to individual workers to decide how to do it." (Respondent 17, Midwife, 6 years)

"We don't have a formal policy for measuring blood loss, so it's not prioritized by the hospital management. If they don't enforce it, we can't take it seriously." (Respondent 13, Nurse, 7 years)

"There's no support from management to ensure we have the right tools or training. It's a problem when leadership doesn't push for better practices." (Respondent 5, Nurse, 6 years)

"The lack of clear policies means there's no accountability for blood loss measurement. It's not integrated into the hospital's routine care protocols." (Respondent 8, Midwife, 6 years)

"Management doesn't give us the resources we need for accurate blood loss measurement. There's no follow-up to see if we're using the tools correctly." (Respondent 10, Nurse, 7 years)

Subtheme 5.2: Lack of Integration of Tool Use into Standard Protocols

Another issue highlighted by respondents was the lack of integration of tool use into standard care protocols. Many participants reported that blood loss measurement was not a routine part of the clinical guidelines and was often treated as an optional task. The absence of systematic integration into the patient care process meant that tools were often underutilized or not prioritized. Respondents stressed the importance of formalizing the use of measurement tools in clinical protocols to ensure that healthcare workers consistently use them in postpartum hemorrhage cases.

"We don't have standard protocols for measuring blood loss. It's not something that is regularly emphasized in our clinical guidelines." (Respondent 2, Nurse, 7 years)

"If the use of these tools was a part of our standard procedures, it would be easier to follow and make sure everyone is on the same page." (Respondent 9, Midwife, 6 years)

"There is no official protocol for measuring blood loss. We just do it as we see fit, which isn't always accurate." (Respondent 15, Nurse, 6 years)

"If measuring blood loss was embedded in our care standards, we would be more consistent in using the tools correctly." (Respondent 12, Midwife, 5 years)

"Standardizing blood loss measurement in our hospital protocols would make it a priority for everyone involved in patient care." (Respondent 8, Nurse, 6 years)

"We need to incorporate the use of these tools into our daily care protocols. Right now, it's just something we do when we can." (Respondent 7, Midwife, 7 years)

CHAPTER FIVE

DISCUSSION OF FINDINGS

5.0 Introduction

This chapter presents a comprehensive discussion of the findings of the study on the utilization of blood loss measurement tools in the detection of postpartum hemorrhage (PPH) among healthcare workers at Adeoyo Maternity Teaching Hospital, Ibadan. The discussion is organized according to the study objectives and the major themes that emerged through thematic analysis of the qualitative data.

The findings are examined in relation to existing empirical literature on quantitative blood loss (QBL), visual estimation practices, maternal health systems in low-resource settings, and global recommendations for postpartum hemorrhage management. In addition, the discussion reflects on the broader institutional and health system factors influencing clinical practice, particularly within the Nigerian maternal healthcare context.

The findings revealed that although healthcare workers recognize the importance of accurate blood loss measurement in preventing maternal morbidity and mortality, several contextual constraints limit consistent implementation. These constraints include high patient volume, inadequate equipment, financial limitations, insufficient training, staffing shortages, and weak policy enforcement.

At the same time, important facilitators were identified, including mentorship, teamwork, institutional support, professional development opportunities, and positive patient outcomes. These findings suggest that blood loss measurement practices are not determined solely by

clinical knowledge but are deeply shaped by organizational culture, resource availability, leadership support, and workforce capacity.

The results further demonstrate that reliance on visual estimation—particularly during emergencies and high-patient volume periods—remains common practice. This aligns with evidence from low-resource settings where quantitative measurement tools are inconsistently available or insufficiently integrated into routine care protocols. The tension between clinical urgency and measurement accuracy emerged as a central issue influencing practice.

The discussion highlights areas of convergence and divergence between the present findings and existing studies from Nigeria, sub-Saharan Africa, and other low- and middle-income countries. It also underscores the need for structural and policy-level interventions to strengthen quantitative blood loss measurement practices and improve postpartum hemorrhage outcomes.

Through this analysis, the chapter situates the study findings within broader global efforts to reduce maternal mortality, emphasizing that improving blood loss measurement requires not only tools, but also system-level commitment, workforce strengthening, and institutional accountability.

5.1 Discussion Of Findings

5.1.1 Demographic Characteristics of the Participants

The demographic characteristics of participants in this study reveal a workforce largely composed of nurses and midwives with varying years of clinical experience, ranging from early-career practitioners to those with over a decade of service in maternal health. This distribution is consistent with the staffing profile of many secondary and tertiary maternity

facilities in Nigeria and other sub-Saharan African countries, where nurses and midwives constitute the backbone of obstetric care and are primarily responsible for intrapartum monitoring and postpartum hemorrhage (PPH) detection.

The predominance of nurses and midwives in this study aligns with global maternal health workforce data indicating that midwives deliver the majority of essential maternal and newborn health services in low- and middle-income countries. The World Health Organization (2023) emphasizes that midwives and nurses are central to early detection and management of PPH, particularly in resource-constrained settings where obstetricians may not be immediately available. Similarly, Ibrahim, Ogaji, and Amadi (2024) found that maternal healthcare delivery in primary and secondary facilities in Nigeria relies heavily on midwives and experienced nurses, whose competence significantly influences clinical outcomes.

The variation in years of professional experience among participants is an important finding. In the present study, participants ranged from approximately three to over ten years of experience. This mix of junior and senior staff reflects typical staffing structures in Nigerian public hospitals, where workforce retention challenges coexist with periodic recruitment. Mukhtar, Nasir, and Aliyu (2024) reported that workforce retention among healthcare professionals in Nigeria is influenced by institutional support, marital stability, and working conditions, resulting in heterogeneous experience levels within facilities. Such diversity in experience can have both positive and negative implications for clinical practice.

On one hand, experienced staff often serve as informal mentors, contributing to skill transfer and clinical confidence among junior workers. This finding in the present study resonates with Ibrahim et al. (2024), who demonstrated that experience level significantly affects

leadership roles, mentorship capacity, and decision-making in maternal health settings. On the other hand, facilities with a high proportion of less-experienced staff may face challenges in implementing evidence-based practices such as quantitative blood loss (QBL) measurement, especially when structured mentorship systems are absent.

The demographic composition of this study also reflects gendered patterns typical of maternal health services. Although gender was not the central focus of this study, nursing and midwifery in Nigeria remain predominantly female professions. This aligns with broader global workforce analyses indicating that maternal health services are largely delivered by women (World Health Organization, 2023). Workforce gender dynamics can influence communication patterns, teamwork structures, and professional hierarchies, which in turn may shape adherence to clinical protocols such as standardized blood loss measurement.

Another important demographic aspect is years of service within the same facility. Research suggests that institutional familiarity may influence compliance with protocols and adaptation to organizational culture. Ekpo, Fajemirokun, and Ibrahim (2021) found that healthcare workers who had spent longer periods in a facility were more likely to adopt or resist innovations depending on prevailing institutional norms. In contexts where visual estimation has historically been the dominant practice, experienced staff may either champion change toward quantitative methods or reinforce entrenched habits.

Furthermore, the presence of nurse-midwives among participants reflects the dual clinical responsibilities common in Nigerian maternity hospitals. Nurse-midwives often manage both routine deliveries and obstetric emergencies, positioning them at the frontline of PPH detection. Gyamfi, Boateng, and Akuffo (2022), in a study conducted in Ghana, observed that midwives' clinical exposure and years of practice significantly influenced their perceptions of

blood loss measurement accuracy and their confidence in using standardized tools. The current study mirrors this pattern, as participants with longer experience often expressed stronger opinions about both the strengths and limitations of available measurement methods.

The demographic findings of this study also intersect with training exposure. Several participants indicated that formal training on quantitative blood loss measurement was limited or irregular. Evidence from structured training interventions demonstrates that demographic factors such as professional cadre and years of experience influence training uptake and retention. Okafor, Onwudiegwu, and Ukwuani (2022) reported that structured training programs in Nigerian primary healthcare centers significantly improved blood loss measurement accuracy, particularly among midwives with fewer than ten years of experience. Similarly, Ngugi, Nyamai, and Kabiru (2020) found that simulation-based training improved competence across experience levels, although younger practitioners demonstrated faster adoption of standardized protocols.

Comparatively, findings from other low-resource settings support the idea that demographic characteristics influence implementation of quantitative blood loss measurement. Garin, Tiwari, and Kamble (2022) observed in rural India that younger healthcare workers were more receptive to adopting calibrated drapes and digital tools, whereas older practitioners were more reliant on visual estimation due to long-standing habits. The current study reflects similar patterns, where reliance on visual estimation was often normalized among more experienced staff during high-pressure situations.

In contrast, evidence from higher-resource settings suggests that demographic variability has less impact where institutional protocols are strongly enforced. Althaus, Bates, and Danilenko-Dixon (2022) found that in U.S. maternity units where quantitative blood loss

measurement is mandated, adherence did not significantly differ by years of experience, suggesting that strong institutional systems can mitigate individual-level variability. The contrast underscores the importance of systemic enforcement and structured implementation strategies in shaping practice, beyond individual demographic characteristics.

The demographic profile of participants in this study reflects the realities of maternal healthcare staffing in Nigeria and similar contexts. The mixture of experience levels, the dominance of nurses and midwives, and the presence of informal mentorship structures are consistent with contemporary literature from sub-Saharan Africa and other low- and middle-income countries. These characteristics form the foundation upon which both barriers and facilitators to blood loss measurement are constructed.

The demographic findings therefore provide essential context for understanding subsequent themes in this discussion, particularly those related to training gaps, mentorship, workload, and institutional support. They reinforce the argument that effective implementation of quantitative blood loss measurement must account for workforce composition, professional experience, and structured capacity-building strategies within maternal healthcare systems.

5.1.2 The availability of blood loss measurement tools in postpartum haemorrhage detection among health workers

The findings of this study revealed that the availability of blood loss measurement tools in Adeoyo Maternity Teaching Hospital was inconsistent and often inadequate, particularly during high-demand periods. Although some essential tools such as gauze, suction machines, and calibrated jars were reportedly present, participants consistently described shortages, limited quantities, malfunctioning equipment, and the complete absence of blood collection

drapes. These constraints significantly influenced their reliance on visual estimation during postpartum haemorrhage (PPH) management.

This finding aligns with growing evidence across sub-Saharan Africa that limited access to standardized quantitative blood loss (QBL) tools remains a major barrier to effective PPH detection. Ekpo, Fajemirokun, and Ibrahim (2021) reported that Nigerian primary healthcare centers frequently lacked calibrated drapes and functional measurement equipment, resulting in inconsistent implementation of QBL protocols. Similarly, Olatunji, Owoaje, and Alabi (2021) found that while some tertiary facilities in Nigeria possessed basic tools, inadequate supply and poor maintenance reduced their routine use, reinforcing dependence on visual estimation.

The absence of blood drapes identified in the present study is particularly significant. Blood collection drapes have been shown to improve the accuracy of postpartum blood loss quantification in low-resource settings. Fawole, Awolude, Adeniji, and Onafowokan (2021) demonstrated in a Nigerian intervention study that the introduction of calibrated drapes significantly improved early detection of PPH compared to visual estimation alone. Likewise, Mwangi, Ngugi, and Nyamai (2021) reported in Kenya that maternity units using blood collection drapes achieved more accurate documentation of blood loss and earlier clinical intervention. The contrast between these documented benefits and the reported absence of drapes in the current study underscores a critical gap between evidence-based recommendations and real-world implementation.

The findings also highlighted limited numbers of suction machines and calibrated jars, often shared across wards or departments. This mirrors findings from Garin, Tiwari, and Kamble (2022), who observed in rural India that healthcare workers frequently had to wait for

equipment to become available, leading to delayed or omitted measurement. In their study, resource competition during peak service hours significantly reduced adherence to QBL protocols. A similar pattern was evident in the present study, where high patient volume compounded equipment scarcity.

Globally, authoritative guidelines emphasize that availability of standardized measurement tools is foundational to effective PPH management. The World Health Organization (2023) explicitly recommends routine quantitative assessment of blood loss using calibrated collection devices or gravimetric methods. However, the WHO also acknowledges that implementation in low- and middle-income countries is often constrained by procurement challenges, supply chain weaknesses, and limited institutional funding. The experiences of participants in this study reflect these systemic limitations.

Financial constraints were repeatedly linked to inadequate equipment supply. Evidence from Nigerian healthcare settings supports this connection. Odetola and Oladimeji (2021) found that insufficient institutional funding directly limited procurement of obstetric emergency equipment, including calibrated jars and blood measurement tools. Similarly, Musa, Adeyemo, and Olatunji (2020) noted that although electronic and digital blood loss measurement technologies were introduced in some urban hospitals in Nigeria, sustainability was hindered by maintenance costs and inconsistent budget allocation. The current study reinforces the argument that equipment availability is inseparable from broader health financing structures.

Another important dimension emerging from this study is the issue of functional availability. Participants reported that even when equipment existed, it was sometimes broken, poorly maintained, or lacking clear calibration markings. This finding is consistent with reports from

Ethiopia and Ghana. Kebede, Worku, and Tessema (2021) found that malfunctioning or poorly calibrated equipment reduced healthcare workers' trust in measurement tools, increasing reliance on subjective estimation. Similarly, Gyamfi, Boateng, and Akuffo (2022) observed that healthcare workers in Ghana expressed frustration when available tools were unreliable, leading to inconsistent use. The present study echoes these concerns, demonstrating that mere presence of tools does not guarantee effective utilization.

Comparatively, high-income settings where QBL measurement is institutionalized demonstrate a different pattern. Althaus, Bates, and Danilenko-Dixon (2022) reported that in facilities where gravimetric measurement was standardized and adequately resourced, compliance rates were significantly higher and visual estimation was rarely relied upon exclusively. The contrast highlights how structural investment and policy enforcement shape clinical behavior.

Furthermore, the findings suggest that availability alone is insufficient without integration into routine protocols. Even when some tools were accessible, their inconsistent supply during emergencies meant that measurement practices were not standardized. This resonates with findings by Akinyemi, Olubunmi, and Balogun (2023), who demonstrated that structured obstetric hemorrhage kits improved measurement consistency only when adequately stocked and embedded into institutional practice guidelines. Where supply chains were weak, the kits' effectiveness diminished.

The findings of this study strongly corroborate existing literature indicating that inadequate availability, insufficient quantity, and limited functionality of blood loss measurement tools are persistent barriers to accurate PPH detection in low-resource settings. The evidence consistently shows that when calibrated drapes, gravimetric tools, and functional suction

devices are reliably available, accuracy improves and maternal outcomes are enhanced. Conversely, when supply is inconsistent, visual estimation becomes normalized, increasing the risk of delayed intervention.

Therefore, the availability of blood loss measurement tools must be understood not merely as a logistical issue but as a systemic determinant of maternal health outcomes. Sustainable procurement, maintenance systems, and integration into standardized protocols are essential for translating global recommendations into effective clinical practice in settings such as Adeoyo Maternity Teaching Hospital.

5.1.3 The use of blood loss measurement tools in postpartum haemorrhage detection among health workers

The findings of this study indicate that although blood loss measurement tools such as gauze, suction machines, and calibrated jars were known to healthcare workers, their use in postpartum haemorrhage (PPH) detection was inconsistent and highly dependent on context. Participants reported that visual estimation remained common practice, particularly during emergencies, high patient volume, and periods of equipment shortage. Quantitative methods were more likely to be used when resources were available and when cases were perceived as severe or high risk. This pattern reflects a hybrid practice environment in which objective measurement coexists with subjective estimation.

This finding is consistent with contemporary literature from low- and middle-income countries, which shows that despite global recommendations supporting quantitative blood loss (QBL) assessment, visual estimation remains prevalent. Olatunji, Owoaje, and Alabi (2021) found in Nigerian hospitals that visual estimation was still widely practiced even where calibrated tools were present, largely due to workload pressures and ingrained clinical

habits. Similarly, Garin, Tiwari, and Kamble (2022) reported that healthcare workers in rural India frequently defaulted to visual estimation during high-pressure situations, citing time constraints and workflow disruption as key barriers to routine QBL use.

The continued reliance on visual estimation observed in this study is concerning given substantial evidence that visual methods underestimate blood loss. Althaus, Bates, and Danilenko-Dixon (2022) demonstrated that gravimetric measurement provides significantly greater accuracy compared to visual estimation in postpartum patients. Their findings align with earlier global evidence showing systematic underestimation when blood loss exceeds 500 mL, which can delay recognition of PPH and timely intervention. The present study's findings suggest that similar risks may exist within the study setting, particularly during busy shifts when estimation replaces measurement.

However, participants also acknowledged the clinical importance of using objective tools when available. This reflects awareness of evolving global standards. The World Health Organization (2023) recommends routine quantitative assessment of postpartum blood loss using calibrated drapes, collection containers, or gravimetric methods. In facilities where QBL has been institutionalized, adherence improves detection rates and documentation accuracy. For instance, Fawole, Awolude, Adeniji, and Onafowokan (2021) found that introducing calibrated drapes in Nigerian facilities significantly increased the identification of PPH cases compared to visual assessment alone. This demonstrates that when tools are actively used, clinical outcomes can improve.

The selective use of measurement tools observed in this study also mirrors findings from Ghana. Gyamfi, Boateng, and Akuffo (2022) reported that midwives recognized the superiority of quantitative methods but admitted to inconsistent application due to workflow

challenges. In their study, health workers described QBL as ideal in theory but difficult to sustain during emergencies. The present findings resonate strongly with this perspective, as participants emphasized the tension between rapid response and measurement precision.

Another important aspect of tool use identified in this study is situational prioritization. Participants reported that calibrated jars and suction devices were more likely to be used in surgical cases or severe bleeding scenarios, while routine vaginal deliveries were often managed using visual estimation. This selective application reflects a risk-based approach rather than a standardized protocol-driven one. Kebede, Worku, and Tessema (2021), in a systematic review of low-resource settings, observed similar patterns, noting that QBL methods were frequently reserved for complicated cases rather than implemented universally. This selective practice undermines early detection, as PPH can occur unpredictably even in low-risk deliveries.

Training has also been shown to influence the consistency of tool use. Okafor, Onwudiegwu, and Ukwuani (2022) demonstrated that structured training interventions in Nigerian primary healthcare centers significantly improved routine application of quantitative measurement tools. Facilities that incorporated training alongside protocol enforcement achieved more consistent use. The inconsistency observed in the present study may therefore be linked not only to resource constraints but also to gaps in structured reinforcement mechanisms.

Comparatively, in higher-resource contexts where QBL is embedded into mandatory clinical protocols, variability in tool use is markedly reduced. Althaus et al. (2022) reported high compliance with gravimetric measurement in U.S. maternity units following institutional mandates. This contrast highlights the importance of integrating measurement tools into standard operating procedures rather than leaving their use to individual discretion.

Additionally, technological innovations have shown promise in improving tool utilization. Bashir, Ranjan, and Sharma (2023) reported that mobile application–supported blood loss tracking improved accuracy and consistency among rural healthcare workers in India. Similarly, Akinyemi, Olubunmi, and Balogun (2023) found that obstetric hemorrhage kits increased measurement reliability when consistently deployed. These studies suggest that structured systems and supportive innovations can enhance sustained use of blood loss measurement tools, even in resource-constrained environments.

The findings of this study therefore reflect a transitional stage in the adoption of quantitative blood loss practices. Health workers demonstrate awareness of the importance of accurate measurement and use tools when feasible, yet structural, workload, and systemic constraints limit universal implementation. This partial adoption creates variability in practice, increasing the risk of delayed PPH detection during high-demand periods.

The study's findings align with regional and global literature indicating that knowledge of quantitative blood loss methods does not automatically translate into consistent practice. Sustainable integration requires adequate resources, structured protocols, continuous training, and institutional accountability. Without these systemic supports, visual estimation is likely to persist as the default method, particularly in high-pressure clinical environments.

5.1.4 The facilitators of the use of blood loss measurement tools in postpartum hemorhhage detection among health workers

The findings of this study identified several important facilitators influencing the use of blood loss measurement tools in postpartum haemorrhage (PPH) detection. These facilitators included support from doctors and senior midwives, teamwork and collaboration, institutional provision of equipment, continuous professional development, mentorship and peer learning,

and motivation derived from positive patient outcomes. Collectively, these elements highlight that successful implementation of quantitative blood loss (QBL) measurement is not solely dependent on tool availability but is strongly shaped by organizational culture, leadership, and professional capacity.

One of the strongest facilitators identified in this study was support from senior healthcare providers, particularly obstetricians and experienced midwives. Participants reported that when doctors emphasized the importance of accurate blood loss measurement and actively supervised practice, adherence improved. This finding aligns with evidence from Ghana, where Gyamfi, Boateng, and Akuffo (2022) found that leadership involvement significantly influenced compliance with PPH management protocols. In their study, facilities where obstetricians actively reinforced quantitative measurement practices demonstrated higher consistency in tool use. This suggests that clinical authority and role modeling play critical roles in shaping practice behaviors.

Collaboration and teamwork also emerged as key enablers in the present study. Participants emphasized that effective communication between nurses, midwives, anesthetists, and doctors enhanced timely and accurate blood loss assessment. Similar findings were reported by Bashir, Patel, and Yadav (2021), who found that team-based training interventions significantly improved measurement accuracy in public hospitals. Their study demonstrated that multidisciplinary simulations strengthened coordination during obstetric emergencies and reduced reliance on visual estimation. The present findings therefore support the broader literature indicating that interprofessional collaboration enhances adherence to standardized obstetric protocols.

Institutional support and resource provision were also identified as facilitators. Participants noted that when management ensured the availability of calibrated jars, suction machines, and adequate supplies, they were more likely to use measurement tools consistently. This is consistent with findings by Ekpo, Fajemirokun, and Ibrahim (2021), who reported that institutional commitment to procuring and maintaining QBL equipment significantly improved implementation in Nigerian healthcare settings. Similarly, Akinyemi, Olubunmi, and Balogun (2023) demonstrated that the introduction of structured obstetric hemorrhage kits improved measurement reliability when facilities ensured sustained supply and oversight. These findings suggest that organizational investment and logistical support directly influence clinical practice.

Continuous education and professional development were also identified as critical facilitators in this study. Participants who had attended workshops or refresher courses reported greater confidence and competence in using measurement tools. This aligns with evidence from Okafor, Onwudiegwu, and Ukwuani (2022), who found that structured training programs in Nigerian primary healthcare centers significantly improved quantitative measurement accuracy. Similarly, Ngugi, Nyamai, and Kabiru (2020) demonstrated that simulation-based training in Kenya enhanced both technical skills and confidence among maternity staff. These studies collectively confirm that training strengthens the practical application of QBL methods and reduces dependence on subjective estimation.

Mentorship and peer learning emerged as additional facilitators. Participants described learning from experienced colleagues through hands-on guidance during real cases. This informal knowledge transfer appears particularly important in settings where structured training is irregular. Ibrahim, Ogaji, and Amadi (2024) found that mentorship within maternal

healthcare facilities enhances leadership development and skill acquisition, particularly among junior staff. Their findings support the present study's observation that mentorship fosters confidence and promotes adherence to best practices. However, they also caution that informal mentorship must be supported by institutional structures to ensure consistency and sustainability.

Another significant facilitator identified was the motivation derived from positive patient outcomes. Participants expressed that witnessing improved maternal outcomes following accurate blood loss measurement reinforced their commitment to using the tools. This experiential reinforcement is supported by evidence from Fawole, Awolude, Adeniji, and Onafowokan (2021), who reported improved PPH detection rates and earlier interventions following implementation of calibrated drapes in Nigerian facilities. Observing tangible improvements in patient recovery strengthened staff acceptance of quantitative methods. Similarly, Mwangi, Ngugi, and Nyamai (2021) found that improved documentation and early detection outcomes in Kenyan maternity units encouraged continued use of blood collection drapes. These findings suggest that visible clinical benefits enhance behavioral adoption of new practices.

Leadership communication also played a facilitative role in this study. Participants indicated that when hospital management clearly communicated the importance of measurement tools and integrated them into expectations for practice, staff compliance increased. The World Health Organization (2023) emphasizes that institutional leadership and policy reinforcement are essential for effective implementation of PPH prevention strategies. Facilities that embed QBL measurement into routine standards and audit mechanisms demonstrate higher

compliance rates. The present findings reinforce the importance of strong managerial advocacy in shaping clinical norms.

Comparatively, studies from higher-resource settings demonstrate that facilitators such as standardized protocols, mandatory documentation systems, and continuous audits significantly enhance QBL adoption. Althaus, Bates, and Danilenko-Dixon (2022) observed high compliance rates in facilities where quantitative measurement was institutionalized through policy mandates and regular performance reviews. The contrast highlights that while individual motivation and mentorship are important, systemic reinforcement is critical for sustained practice change.

The facilitators identified in this study reflect a multifaceted interplay between leadership, teamwork, institutional support, training, mentorship, and positive clinical outcomes. These findings are strongly supported by recent peer-reviewed literature from Nigeria and other low-resource settings. They underscore that improving blood loss measurement practices requires a supportive ecosystem rather than isolated interventions.

The evidence suggests that when healthcare workers are adequately trained, supported by senior colleagues, provided with functional equipment, and motivated by visible patient benefits, adherence to quantitative blood loss measurement significantly improves. Therefore, strengthening these facilitators within maternity units can play a crucial role in enhancing PPH detection accuracy and reducing maternal morbidity and mortality.

5.1.5 The barriers that hinder blood loss measurement tools in postpartum hemorrhage detection among health workers

The findings of this study revealed multiple interrelated barriers hindering the effective use of blood loss measurement tools in postpartum haemorrhage (PPH) detection. These barriers included inadequate resources and equipment, financial constraints, lack of training and knowledge, high workload and staffing shortages, and institutional and systemic weaknesses such as poor policy enforcement and weak managerial support. The convergence of these factors created an environment in which visual estimation frequently replaced standardized quantitative blood loss (QBL) measurement.

One of the most prominent barriers identified was inadequate availability and functionality of essential tools such as suction machines, calibrated jars, and blood collection drapes. Participants described frequent shortages, shared equipment between wards, and malfunctioning devices. This finding aligns with Ekpo, Fajemirokun, and Ibrahim (2021), who reported that limited supply and poor maintenance of measurement equipment significantly reduced adherence to QBL protocols in Nigerian healthcare settings. Similarly, Garin, Tiwari, and Kamble (2022) found in rural India that shortages of calibrated drapes and unreliable equipment forced healthcare workers to revert to visual estimation, particularly during emergencies. These studies confirm that logistical limitations remain a primary barrier to accurate PPH detection in low-resource settings.

Financial constraints emerged as another major barrier in the present study. Participants indicated that insufficient institutional funding delayed procurement, maintenance, and replacement of measurement tools. In some cases, patients were expected to provide basic supplies such as gauze and drapes, further complicating accurate assessment. Odetola and

Oladimeji (2021) similarly identified funding limitations as a critical obstacle to effective PPH management in Nigerian primary healthcare centers. Their findings emphasized that inadequate budget allocation for obstetric emergency equipment undermines implementation of evidence-based practices. At a broader level, the World Health Organization (2023) acknowledges that health financing gaps in low- and middle-income countries hinder full implementation of recommended PPH assessment strategies, including quantitative blood loss measurement.

Lack of training and insufficient knowledge also significantly hindered tool utilization in this study. Some participants reported limited exposure to formal instruction on calibration procedures and proper gravimetric techniques. Others indicated that training sessions were infrequent or predominantly theoretical. These findings are consistent with Ngwenya, Moyo, and Tshimanga (2020), who documented substantial training gaps in blood loss assessment across low-resource settings, leading to persistent reliance on visual estimation. Odetola and Oladimeji (2021) further observed that limited knowledge regarding the interpretation and application of measurement tools contributed to inconsistent use in Nigerian facilities. The persistence of knowledge gaps undermines the transition from subjective to objective assessment methods.

Workload and staffing shortages were also critical barriers identified in the study. Participants described high patient-to-staff ratios, particularly during peak service periods, which limited the time available for careful quantitative measurement. Under such conditions, rapid visual estimation was often perceived as the only practical option. This finding is strongly supported by research from Ghana and India. Gyamfi, Boateng, and Akuffo (2022) reported that heavy workload and understaffing reduced compliance with standardized PPH

protocols. Likewise, Garin et al. (2022) noted that time pressure during emergencies significantly reduced the likelihood of performing gravimetric measurement. The present findings reinforce the understanding that accurate measurement is not merely a technical issue but is deeply influenced by workforce capacity and service demand.

Institutional and systemic challenges further compounded these barriers. Participants indicated that blood loss measurement was not fully integrated into routine protocols and that policy enforcement was weak. In the absence of clear mandates and routine audits, tool use became discretionary rather than standardized. Kebede, Worku, and Tessema (2021) highlighted in their systematic review that inconsistent policy integration across low-resource settings contributes to variability in PPH detection practices. Facilities lacking formal guidelines and accountability mechanisms were more likely to depend on individual clinician judgment rather than standardized measurement.

The persistence of visual estimation, despite global recommendations favoring quantitative methods, illustrates the cumulative effect of these barriers. Althaus, Bates, and Danilenko-Dixon (2022) demonstrated that in settings where QBL measurement was mandated and supported by structured protocols, adherence improved significantly. The contrast between such environments and the context of the present study underscores how systemic support mitigates individual-level and logistical barriers.

Furthermore, cultural and habitual practice patterns may reinforce resistance to change. Olatunji, Owoaje, and Alabi (2021) found that long-standing reliance on visual estimation in Nigerian hospitals created entrenched behavioral norms that were difficult to alter without comprehensive institutional reform. Even when tools were available, some practitioners

reverted to estimation due to familiarity and perceived efficiency. The present study reflects similar tendencies, particularly during high-pressure scenarios.

Taken together, the barriers identified in this study are consistent with contemporary literature from Nigeria and other low- and middle-income countries. They reveal that effective implementation of quantitative blood loss measurement requires more than the physical presence of tools. It demands adequate financing, workforce strengthening, continuous professional development, clear institutional protocols, and leadership enforcement. Without these systemic supports, healthcare workers are compelled to rely on subjective methods, increasing the risk of delayed PPH recognition and adverse maternal outcomes.

The findings therefore highlight the need for comprehensive health system strengthening interventions that address structural, educational, and managerial gaps simultaneously. Addressing these barriers is essential to aligning clinical practice with global standards and improving maternal survival outcomes in settings such as Adeoyo Maternity Teaching Hospital.

5.2 Implication of Findings to Nursing

The findings of this study have significant implications for nursing practice, nursing education, nursing leadership, and health policy, particularly within maternal and child health services in low-resource settings. Given that nurses and midwives constitute the primary workforce responsible for intrapartum monitoring and postpartum haemorrhage (PPH) detection, strengthening evidence-based blood loss measurement practices is fundamentally a nursing priority.

One major implication for nursing practice is the urgent need to transition from predominant reliance on visual estimation to routine quantitative blood loss (QBL) measurement. The study findings revealed that visual estimation remains common, especially during high workload periods and emergencies. However, strong evidence demonstrates that visual estimation consistently underestimates blood loss, particularly when volumes exceed 500 mL. Althaus, Bates, and Danilenko-Dixon (2022) found that gravimetric measurement significantly improves accuracy compared to visual estimation in postpartum patients. Similarly, Kebede, Worku, and Tessema (2021), in a systematic review, concluded that quantitative methods enhance early detection of PPH and reduce the risk of delayed intervention. For nursing practice, this means that accurate blood loss measurement should be embedded as a routine, non-optional component of postpartum monitoring.

The findings also imply that nursing leadership must play a central role in advocating for institutional integration of QBL into standard care protocols. Participants in this study indicated that tool use was inconsistent in the absence of clear policies and enforcement mechanisms. Evidence from Ekpo, Fajemirokun, and Ibrahim (2021) demonstrates that institutional support and policy enforcement significantly improve implementation of quantitative measurement methods in Nigerian healthcare settings. Therefore, nurse managers and senior midwives should champion protocol standardization, ensuring that blood loss measurement is documented systematically in delivery records and audited regularly.

Another critical implication concerns continuous professional development. The study revealed training gaps and irregular refresher programs, which contributed to inconsistent tool usage. Evidence-based interventions show that structured training improves competence and adherence. Okafor, Onwudiegwu, and Ukwuani (2022) reported that structured training

programs in Nigerian primary healthcare centers significantly improved blood loss measurement accuracy. Likewise, Ngugi, Nyamai, and Kabiru (2020) found that simulation-based training enhanced technical skills and confidence among maternity staff in Kenya. For nursing education and practice, this underscores the need to incorporate regular competency-based workshops, simulation drills, and hands-on training in PPH management.

The findings further suggest that mentorship and peer learning should be formalized within nursing structures. Participants identified mentorship as a facilitator of skill acquisition, yet it remained informal and inconsistent. Ibrahim, Ogaji, and Amadi (2024) emphasize that structured mentorship programs in maternal healthcare improve leadership development, confidence, and adherence to best practices. Nursing departments should therefore institutionalize mentorship systems, pairing junior nurses with experienced practitioners to strengthen practical competence in QBL techniques.

Workload and staffing shortages identified in the study also have direct implications for nursing workforce planning. High patient-to-nurse ratios limited the ability to perform accurate measurements. Research consistently demonstrates that inadequate staffing compromises patient safety. Gyamfi, Boateng, and Akuffo (2022) observed that heavy workload reduced adherence to PPH management protocols in Ghanaian facilities. For nursing administration, this highlights the importance of advocating for adequate staffing levels to ensure safe maternal care. Workforce strengthening must be recognized as a patient safety strategy rather than solely an administrative concern.

Resource availability is another critical nursing implication. The study revealed shortages of calibrated jars, suction machines, and blood collection drapes. Evidence shows that when standardized tools are consistently available, detection improves. Fawole, Awolude, Adeniji,

and Onafowokan (2021) demonstrated that the introduction of calibrated drapes significantly improved PPH detection in Nigerian facilities. Nursing leaders must therefore collaborate with hospital management and policymakers to ensure procurement, maintenance, and sustained supply of essential measurement tools.

Additionally, the findings imply that documentation practices require strengthening. Accurate measurement must be accompanied by proper recording to inform clinical decision-making. The World Health Organization (2023) emphasizes routine documentation of quantified blood loss as part of comprehensive PPH prevention strategies. Nurses, being responsible for most intrapartum documentation, must be equipped and empowered to maintain accurate records that support early intervention.

From a broader evidence-based practice perspective, the findings reinforce the need for integrating research into routine nursing care. Adoption of QBL aligns with global best practices and maternal mortality reduction strategies. According to the World Health Organization (2023), postpartum haemorrhage remains a leading cause of maternal mortality globally, particularly in low-resource settings. Nursing practice grounded in evidence-based measurement techniques directly contributes to achieving Sustainable Development Goal targets related to maternal health.

Finally, the findings suggest that nursing advocacy should extend beyond individual facilities to policy engagement. Collaboration with government agencies and non-governmental organizations is essential to secure funding, strengthen supply chains, and support workforce training. Evidence from Akinyemi, Olubunmi, and Balogun (2023) shows that structured obstetric hemorrhage kits improved measurement accuracy when supported institutionally.

Nurses, particularly those in leadership positions, must participate in policy discussions to ensure that maternal health priorities receive adequate attention and funding.

In summary, the implications of this study to nursing are multidimensional. They include strengthening routine quantitative blood loss measurement, institutionalizing training and mentorship, advocating for adequate staffing and resources, reinforcing documentation standards, and promoting evidence-based maternal health policies. Implementing these measures will not only improve clinical accuracy but also enhance maternal safety outcomes in settings similar to Adeoyo Maternity Teaching Hospital.

5.3 Limitations of the Study

Despite the important insights generated by this study on the utilization of blood loss measurement tools in postpartum haemorrhage (PPH) detection, several limitations must be acknowledged. These limitations relate to study design, sampling, data collection methods, contextual scope, and generalizability.

First, the study was conducted in a single tertiary maternity facility, Adeoyo Maternity Teaching Hospital, Ibadan. While the hospital provides a high volume of obstetric services and reflects realities common to many public healthcare institutions in Nigeria, findings may not be fully generalizable to other settings such as private hospitals, rural primary healthcare centers, or facilities in other geopolitical zones. Health system capacity, staffing patterns, funding structures, and administrative support vary significantly across institutions. Similar contextual limitations have been noted in qualitative maternal health studies conducted in Nigeria and other sub-Saharan African countries, where facility-specific factors strongly influence implementation outcomes (Ekpo, Fajemirokun, & Ibrahim, 2021; Gyamfi, Boateng, & Akuffo, 2022).

Second, the qualitative design of the study, while appropriate for exploring perceptions, experiences, and contextual factors, limits the ability to quantify the magnitude of tool utilization or measure the statistical association between barriers and clinical outcomes. The findings reflect subjective experiences and self-reported practices rather than direct observational data or documented clinical audits. Self-reported practices may be influenced by recall bias or social desirability bias, particularly when participants are discussing adherence to recommended guidelines. Research on implementation of quantitative blood loss (QBL) has shown that discrepancies may exist between reported and actual practice when compared to audit data (Althaus, Bates, & Danilenko-Dixon, 2022).

Third, the study focused primarily on nurses and midwives and did not extensively include obstetricians, anesthetists, or hospital administrators as primary participants. Although nurses and midwives are central to PPH detection and monitoring, blood loss measurement is often a multidisciplinary responsibility. Excluding broader managerial and physician perspectives may limit understanding of institutional decision-making processes and policy-level barriers. Evidence suggests that leadership engagement across professional cadres significantly influences successful QBL implementation (World Health Organization [WHO], 2023).

Another limitation relates to the absence of direct clinical outcome measures. The study did not assess maternal morbidity or mortality data before and after the use of blood loss measurement tools. Consequently, while participants reported perceived improvements in patient outcomes when tools were used, the study cannot empirically confirm clinical impact within the facility. Previous intervention-based studies have demonstrated measurable reductions in delayed PPH recognition when quantitative methods are implemented

systematically (Fawole, Awolude, Adeniji, & Onafowokan, 2021), but such outcome comparisons were beyond the scope of this qualitative inquiry.

Additionally, resource and workload challenges described by participants may fluctuate over time due to administrative changes, funding cycles, or policy shifts. The findings therefore represent a snapshot of experiences within the study period rather than a longitudinal assessment. Long-term evaluation would provide stronger insight into sustainability of tool utilization and the impact of reforms.

Finally, although efforts were made to ensure credibility and trustworthiness through careful transcription, thematic analysis, and validation of emerging themes, qualitative interpretation inherently involves researcher subjectivity. While reflexivity and systematic coding procedures were applied, the possibility of interpretive bias cannot be entirely eliminated.

Despite these limitations, the study provides valuable contextual understanding of the complex factors influencing blood loss measurement practices in a high-volume maternity setting. The findings offer important groundwork for future quantitative studies, multicenter research, and intervention-based evaluations aimed at strengthening evidence-based PPH detection practices across diverse healthcare environments.

5.4 Summary of the Study

This study examined the utilization of blood loss measurement tools in the detection of postpartum haemorrhage (PPH) among healthcare workers at Adeoyo Maternity Teaching Hospital, Ibadan. The research was undertaken against the backdrop of persistently high maternal mortality rates in Nigeria, where postpartum haemorrhage remains a leading cause of maternal deaths. The study recognized that accurate and timely measurement of blood loss

is central to early PPH detection and effective intervention, as emphasized in global maternal health guidelines.

The study adopted a qualitative research design to explore the experiences, perceptions, and practices of nurses and midwives regarding blood loss measurement. Through thematic analysis of participant narratives, key themes were identified relating to availability, utilization, facilitators, and barriers to the use of blood loss measurement tools.

Findings revealed that while healthcare workers were aware of the importance of accurate blood loss measurement, visual estimation remained widely practiced, particularly during high-patient volume periods and emergency situations. Participants described relying on visual assessment when time constraints, understaffing, or equipment shortages limited the use of calibrated tools.

Regarding availability, the study found that although some tools such as suction machines, gauze, and calibrated jars were present, their quantity and functionality were often inadequate. Blood collection drapes, which are recommended for quantitative blood loss measurement, were largely unavailable. Limited supply and poor equipment maintenance hindered consistent implementation of standardized measurement methods.

The study further identified facilitators that supported tool utilization. These included mentorship from senior staff, teamwork among healthcare providers, continuous professional development opportunities, institutional encouragement, and positive patient outcomes observed following accurate measurement and early intervention. Participants reported increased confidence and competence when guided by experienced colleagues, highlighting the importance of supportive professional environments.

However, significant barriers were identified. These included inadequate resources and equipment shortages, financial constraints affecting procurement and maintenance, insufficient training opportunities, high patient-to-staff ratios, limited time for proper measurement, and weak integration of blood loss measurement into institutional protocols. These systemic and organizational challenges limited consistent adherence to quantitative measurement practices.

The implications of these findings are substantial for nursing practice, education, and leadership. The study underscores the need for institutionalization of quantitative blood loss measurement as a routine standard of care, strengthening of workforce capacity through continuous training, provision of adequate resources, and enforcement of clear clinical guidelines. Without systemic reforms, healthcare workers may continue to rely on subjective estimation methods, potentially delaying PPH recognition and intervention.

In conclusion, this study demonstrates that improving postpartum haemorrhage detection requires more than awareness of best practices. It demands coordinated efforts involving resource allocation, workforce strengthening, policy enforcement, mentorship, and evidence-based training. Addressing these interconnected factors is essential to enhancing maternal safety outcomes and advancing progress toward maternal mortality reduction targets.

5.5 Conclusion

This study examined the utilization of blood loss measurement tools in the detection of postpartum haemorrhage among healthcare workers at Adeoyo Maternity Teaching Hospital, Ibadan. The findings demonstrate that although healthcare workers recognize the importance of accurate blood loss measurement in preventing maternal morbidity and mortality,

consistent implementation of quantitative methods remains constrained by systemic, institutional, and workforce-related challenges.

The study revealed that visual estimation continues to be widely practiced, particularly during emergencies and periods of high patient volume. While some measurement tools such as suction machines and calibrated jars are available, their limited quantity, inconsistent functionality, and the absence of critical tools such as blood collection drapes reduce the likelihood of routine quantitative measurement. Financial limitations, inadequate staffing, insufficient training opportunities, and weak policy enforcement further compound these challenges.

At the same time, the study identified important facilitators that promote the use of blood loss measurement tools. These include mentorship from experienced colleagues, teamwork and collaboration among healthcare providers, continuous professional development, supportive supervision, and motivation derived from positive patient outcomes. These enabling factors highlight the potential for improvement when institutional support, workforce competence, and leadership commitment are strengthened.

The findings underscore that effective postpartum haemorrhage detection is not solely dependent on individual clinical knowledge but is deeply influenced by structural and organizational factors. Improving maternal outcomes therefore requires a comprehensive approach that integrates adequate resource provision, structured training, standardized clinical protocols, strong managerial support, and sustained workforce development.

In conclusion, strengthening the routine use of blood loss measurement tools in postpartum haemorrhage detection is essential for enhancing patient safety and reducing preventable

maternal complications. Addressing the identified barriers while reinforcing existing facilitators will contribute significantly to improving the quality of maternal healthcare services and advancing efforts toward reducing maternal mortality.

5.6 Recommendation

Based on the findings of this study, several recommendations are proposed to enhance the effective utilization of blood loss measurement tools in postpartum haemorrhage detection among healthcare workers.

1. Hospital management should ensure the consistent availability and adequate supply of essential blood loss measurement tools, including calibrated jars, functional suction machines, blood collection drapes, and weighing scales. Procurement systems should be strengthened to prevent shortages, and regular maintenance schedules should be established to ensure that equipment remains functional and reliable at all times.

2. Healthcare institutions should integrate quantitative blood loss measurement into standard clinical protocols for labour, delivery, and postpartum care. Clear guidelines should be developed and enforced to make blood loss measurement a routine, mandatory component of obstetric practice rather than an optional activity. Regular clinical audits should be conducted to monitor compliance and improve accountability.

3. Hospital management and relevant health authorities should address staffing shortages by recruiting additional nurses and midwives to reduce high patient-to-staff ratios. Adequate staffing will allow healthcare workers sufficient time to perform accurate blood loss measurements without compromising other aspects of patient care.

4. Continuous professional development programs should be organized regularly to enhance the knowledge and skills of healthcare workers in postpartum haemorrhage management. Training should include practical, hands-on sessions on the correct use, calibration, and interpretation of blood loss measurement tools. Simulation-based drills and refresher courses should be conducted periodically to reinforce competence and confidence.

5. A structured mentorship program should be established within maternity units to support less experienced nurses and midwives. Senior staff should be formally assigned mentoring roles to guide junior colleagues in the practical application of quantitative blood loss measurement techniques and evidence-based PPH management.

6. Hospital leadership should strengthen supervisory mechanisms to ensure consistent implementation of measurement practices. Clear communication from management regarding the importance of accurate blood loss measurement for patient safety should be sustained to foster a culture of accountability and quality improvement.

7. Government agencies and non-governmental organizations should collaborate with healthcare institutions to provide financial and material support for maternal health services. Investment in obstetric emergency equipment, training initiatives, and workforce strengthening should be prioritized as part of broader maternal mortality reduction strategies.

8. Finally, further research is recommended to include multi-center studies and quantitative assessments of clinical outcomes associated with the implementation of quantitative

blood loss measurement. Such studies would provide stronger evidence for policy formulation and large-scale adoption of standardized blood loss measurement practices.

Implementing these recommendations will contribute to improving the early detection and management of postpartum haemorrhage, enhancing maternal safety, and strengthening the overall quality of obstetric care.